

For Whom Does Game-Based Sales Training Work?

Expertise-Conditional Effects in an RCT of a Scaffolded
B2B Simulation

Yoshua Yanottama

April 2026

Abstract

Background: Game-based learning (GBL) produces inconsistent outcomes in professional sales training, yet the moderating role of learner expertise remains underexplored.

Methods: A pre-registered randomised controlled trial ($N = 63$) compared a scaffolded B2B sales simulation to a conventional e-learning programme. Sales self-efficacy (SE) and adaptive selling beliefs (ADAPTS) were analysed using ANCOVA with pre-test scores as covariates, following CONSORT recommendations; cognitive load, engagement, and transfer intention were examined with Welch t -tests.

Results: ANCOVA revealed a marginal trend for SE ($F(1,60) = 3.56$, $p = .064$, $\eta_p^2 = .056$, approaching medium effect) and a non-significant difference in ADAPTS ($F(1,60) = 2.22$, $p = .141$). The game condition incurred substantially higher perceived cognitive load ($d = 0.71$) without corresponding gains in germane engagement. An exploratory cluster analysis identified three expertise profiles using pre-intervention features only; novice learners ($n = 10$) showed a large negative effect on adaptive selling gain ($d = -1.88$, $p < .001$), consistent with expertise reversal predictions (Kalyuga et al., 2003), while intermediate and advanced clusters trended positively or neutrally. Within the game group, SE fully mediated the effect of engagement on adaptive selling (indirect effect = 1.10, 95% CI [.43, 2.24]). Expertise moderation was corroborated by differential SE–adaptive selling correlations across conditions ($r = .877$ vs. $.532$, Fisher z , $p = .004$) and by behavioural telemetry (first-attempt latency: $r = -.37$, $p = .031$).

Conclusions: At the aggregate level, the game produced higher cognitive load without superior learning outcomes, constituting a null-to-marginal primary finding. Exploratory analyses provide preliminary evidence that effectiveness is expertise-conditional: novice learners are disadvantaged by the additional interface load imposed before domain schemas exist, while intermediate and advanced learners are neutral or marginally benefited. These exploratory findings require pre-registered replication, but support adaptive deployment strategies—such

as prerequisite screening or dynamic difficulty adjustment—and extend expertise reversal theory to professional GBL contexts as a hypothesis warranting confirmatory investigation.

Keywords: digital game-based learning, cognitive load, randomised controlled trial, expertise reversal, sales self-efficacy, adaptive selling

Contents

Abstract	i
Acknowledgements	vii
1 Introduction	1
1.1 Background	1
1.2 Research Objective	3
1.3 Significance of the Study	3
1.3.1 Theoretical Contributions	4
1.3.2 Practical Contributions	4
1.3.3 Methodological Contributions	4
1.4 Delimitations of the Study	5
2 Literature Review	7
2.1 Sales Education and the Case for DGBL	7
2.1.1 Why DGBL Fits Sales Education	7
2.1.2 Current Approaches to Sales Education	9
2.2 Theoretical and Design Foundations	11
2.2.1 LM–GM Framework	11
2.2.2 Cognitive Load Theory	12

2.2.3	Self-Efficacy	13
2.2.4	Engagement	14
2.2.5	Sales Competency	16
2.3	Theoretical Integration	16
2.3.1	Cognitive Load and Self-Efficacy	16
2.3.2	DGBL as a Moderator of Cognitive Load	17
3	Conceptual Framework and Hypothesis	19
4	Methodology	21
4.1	Research Design	21
4.2	Population	22
4.3	Interventions, Measures, and Procedure	23
4.3.1	Interventions	23
4.3.2	Instruments and Measures	25
4.3.3	Procedure	27
4.4	Data Analysis Plan	29
5	Results	32
5.1	Baseline Equivalence	32
5.2	Hypothesis 1: Between-Group Comparisons	33
5.3	Within-Group Change	36
5.4	Hypothesis 2: Interrelationships Within the Game Group	37
5.5	Behavioral Evidence	39
5.6	Expertise-Conditional Effects	40
6	Discussion	43

6.1	Summary of Findings	43
6.2	Interpretation Through Cognitive Load Theory	44
6.3	Self-Efficacy, Engagement, and Transfer	45
6.4	Expertise-Conditional Effects and Theory Extension	46
6.5	Linkback to the Theoretical Framework	48
6.6	The Importance of Non-Completers	49
6.7	Implications for B2B Sales Training Design	50
6.8	Limitations	51
6.9	Future Research	54
7	Conclusion	56

List of Tables

4.1	Measurement instruments and internal consistency	26
4.2	Hypothesis and Analysis Methods	30
5.1	Baseline equivalence between analyzed groups (0–100 scale)	32
5.2	ANCOVA results for primary outcomes (0–100 scale; $n = 63$)	33
5.3	ITT sensitivity analysis: ANCOVA under three attrition scenarios	34
5.4	Between-group post-test comparisons: secondary outcomes	35
5.5	Within-group pre-post improvement summary (0–100 scale)	36
5.6	Selected correlations within the game group	37
5.7	Condition moderation of construct relationships (Fisher z tests)	38
5.8	Simple slopes: game vs. control advantage at three pre-intervention expertise levels	40
5.9	Cluster profiles: expertise-stratified learner groups ($k = 3$, $n = 63$)	41
5.10	Within-stratum game vs. control: ADAPTS gain by expertise cluster (small- sample corrected effect sizes)	41
6.1	Retrospective power analysis: required N and achieved power for key outcomes	51

Acknowledgements

This manuscript consolidates the finalized statistical findings from the analysis report with the theoretical framing developed in earlier proposal documents. The document is written as a thesis-ready version aligned to the current evidence base.

Chapter 1

Introduction

1.1 Background

Digital game based learning (DGBL) has emerged as a promising design for learning interventions with better learning performance, improved attitude in learning, and higher motivation, in comparison to traditional face-to-face instruction (Chang et al., 2020) or even to online virtual learning environments (Hernández and Moreno, 2019). It has been used in various fields and industries to improve critical thinking and argumentation skills (Noroozi et al., 2020), leadership skills (Sousa and Costa, 2014), and even as part of wider corporate competence development (Senderek et al., 2015). These benefits have led to its increasing acceptance in workplace training, yet barriers persist: lack of available solutions, lack of expertise in authoring, doubts about efficacy (Tay et al., 2022), and the negative social association of games in some settings (Bösche and Kattner, 2011). As Tay et al. suggest, questions of efficacy need to be addressed through stronger alignment with pedagogical principles and teaching strategies, because without a coherent design framework DGBL can suffer from the inclusion of non-essential design and gamification components that potentially hinder the achievement of learning outcomes. Rooting DGBL in sound pedagogical and learning strategies through informed instructional design may improve its appeal in workplace training beyond mere novelty effect and strengthen the validity of claims made about its impact. A useful framework for this intentional design is the learning mechanics–game mechanics framework (LM–GM), which encourages alignment between game mechanics and intended learning outcomes through corresponding learning mechanics (Arnab et al., 2015).

There is growing interest in the potential of DGBL for corporate training, particularly within

the demanding field of sales education (Cummins et al., 2013; Tay et al., 2022), but concerns about efficacy and the attribution of the factors leading to DGBL success cannot be answered adequately by many existing research designs, limiting the explanatory power of current findings and their utility in informing robust DGBL design (All et al., 2021). While many studies cite increased engagement as the key driver of improved learning outcomes over traditional methods (Dahalan et al., 2024; Woo, 2014), the specific factors facilitating that engagement have not been investigated in sufficient depth. Cognitive load theory (Sweller et al., 2011) and self-efficacy theory (Bandura, 1977, 1986) can be leveraged as promising explanatory factors for DGBL effectiveness in sales education and beyond (Barling and Beattie, 1983; Hung et al., 2014).

The DGBL evidence base also suffers from methodological limitations, because the scarcity of randomized controlled trials (RCTs) constrains reliable comparison with traditional designs and limits strong causal inferences regarding effectiveness in promoting learning outcomes (All et al., 2021; Hainey et al., 2016; Tay et al., 2022). All et al.’s (2021) systematic review of DGBL effectiveness assessment identifies several recurring design weaknesses that this study attempts to address: the failure of true random assignment in online recruitment contexts (motivating the use of alternating blocked allocation here), the insufficient attention to instructional time as a factor in attrition (directly evidenced in the present study’s behavioral logs, where the majority of game dropouts occurred at phase transitions associated with the highest time demands), and the under-examined role of motivation to start playing. On the last point, All et al. note that entry motivation is a confound that is rarely measured or controlled; in the present study, it manifests precisely as selective attrition rather than as a baseline variable, allowing the dropout pattern itself to serve as behavioral evidence of the friction that low-entry-motivation participants encounter. These design choices and their consequences are discussed further in Section 4 and Section 5.2. Specifically, there is a need for more empirical investigation into how scaffolding strategies embedded within DGBL for sales training influence the interplay between cognitive load and self-efficacy (Chang et al., 2018; Evans et al., 2024), and how these factors collectively shape sales competency development (Cardinali et al., 2025). While traditional sales education methods face challenges in scalability and in providing safe, repeatable practice (Cummins et al., 2013), there remains a gap in research directly comparing a pedagogically grounded DGBL intervention with standard digital formats in the sales domain. The potential of specific DGBL modalities, such as digital card games (Kordaki and Gousiou, 2017), to address these issues in sales training also warrants closer research attention.

1.2 Research Objective

To guide and focus the investigation, the following research questions are formulated:

- **Research question 1 (RQ1):** How does a scaffolded game-based learning (DGBL) intervention compare to a traditional click-and-continue digital training method in terms of its impact on sales training outcomes?

This primary research question is explored by comparing the two training conditions across several key outcome variables. Specifically, the study examines whether the scaffolded DGBL intervention leads to significantly different levels of learner engagement, sales self-efficacy, perceived cognitive load, adaptive selling, and transfer intention. In addition, it evaluates whether the DGBL group demonstrates stronger improvement on competency-related sales outcomes from pre-test to post-test relative to the traditional digital training group.

- **Research question 2 (RQ2):** Within the context of the scaffolded DGBL sales training, what are the relationships between perceived cognitive load, learner engagement, sales self-efficacy, and sales competency-related development?

This second research question focuses on the internal dynamics within the DGBL intervention group. It investigates the extent to which perceived cognitive load, learner engagement, and sales self-efficacy are interrelated. It also explores how these psychological variables, both individually and collectively, relate to adaptive selling, transfer intention, and broader competency-related sales outcomes. Finally, the study examines whether there is evidence that the influence of managed cognitive load on competency-related outcomes is channeled through self-efficacy and/or learner engagement, thereby clarifying the potential mechanisms at work inside the DGBL environment.

1.3 Significance of the Study

This research is anticipated to make several contributions to the fields of educational technology, instructional design, corporate training, and sales education. Its significance lies in the potential to advance both theoretical understanding and practical application of Digital Game-Based Learning (DGBL).

1.3.1 Theoretical Contributions

The study provides an empirical test of a DGBL intervention designed explicitly around cognitive load theory (CLT), self-efficacy theory, motivation theory, and the structured LM–GM game design framework. The primary aggregate finding — higher cognitive load in the game condition without corresponding gains in self-efficacy or adaptive selling — constitutes a theoretically meaningful result in its own right: it documents a case in which a pedagogically aligned game design imposed extraneous load rather than managing it. The exploratory expertise-conditional pattern (novice disadvantage $d = -1.88$, advanced learner neutrality-to-advantage) provides preliminary evidence that CLT’s expertise reversal mechanism extends to professional DGBL contexts. These findings, taken together, offer a more nuanced account of when and for whom LM–GM-grounded DGBL can be expected to work, while identifying mechanics transparency as a design criterion the framework does not yet operationalise.

1.3.2 Practical Contributions

The study involves the creation and evaluation of a specific scaffolded, card-based digital game for sales training. The aggregate null-to-marginal finding is itself a practically informative result: it establishes that a well-designed, LM–GM-grounded game does not automatically outperform a conventional digital module for a mixed-expertise population. The exploratory expertise-conditional pattern suggests that deployment decisions should account for learner prior knowledge, with novice learners directed to direct instruction before game-based modules are introduced. The high attrition rate in the game condition (52.7%) is an additional practical signal: organizations should anticipate non-trivial dropout rates in game-based online training and treat completion as a deployment outcome in its own right, not only as a statistical filter.

1.3.3 Methodological Contributions

The study makes several methodological contributions to GBL evaluation research. First, it demonstrates the feasibility of conducting a pre-registered randomised controlled trial of a DGBL intervention in a fully online, professional training context, pairing survey-based psychometric outcomes with behavioural telemetry from game logs — a combination that is comparatively rare in the GBL literature (All et al., 2021). Second, the clustering approach

used to operationalise expertise groups was deliberately restricted to pre-intervention features only, avoiding the circularity that arises when post-test or gain variables are included in cluster solutions; this methodological choice is foregrounded as a template for future expertise-stratified GBL analyses. Third, a continuous moderation analysis (pre-outcome $\times$ condition interaction regression) is presented as the primary evidence for expertise moderation, rather than relying solely on the post-hoc cluster solution; this approach avoids the discretisation assumptions inherent in k -means and is more directly replicable in future designs. Fourth, two intention-to-treat sensitivity analyses are reported alongside the per-protocol primary analysis, bounding the range of plausible treatment effects under different attrition assumptions. Fifth, the study contributes a methodological critique that extends beyond this study: immediate post-test comparison is structurally conservative for game-based conditions, because games operate on motivational and schema-building mechanisms whose outputs compound post-training rather than expressing fully at the moment of completion. Adopting delayed retention and transfer endpoints as primary — rather than supplementary — outcomes is proposed as a field-wide standard for comparative GBL research.

In summary, this research aims to bridge theory and practice by developing and evaluating a pedagogically grounded DGBL intervention. It is intended to provide knowledge that can improve the design and implementation of future DGBL initiatives, particularly in the critical area of corporate sales training.

1.4 Delimitations of the Study

This study’s findings should be interpreted within the following scope:

- **Participant Generalizability:** While targeting novice sales professionals, open online recruitment means the sample may be diverse; results primarily apply to this self-selected group.
- **Intervention Specificity:** The study evaluates a specific scaffolded card-based DGBL (“Improve Your Sales Game”); findings may not generalize to all DGBL formats or all sales training contexts.
- **Competency Assessment:** The focus is on foundational sales-process knowledge and scenario-based skills covered in the training, not the full spectrum of sales competencies or actual on-the-job performance.

- **Measurement Timeline:** Outcomes are assessed immediately post-intervention, without long-term follow-up on retention or real-world skill transfer.

Chapter 2

Literature Review

2.1 Sales Education and the Case for DGBL

2.1.1 Why DGBL Fits Sales Education

The evolving landscape of business, the continuous demand for professional upskilling, and the characteristics of modern learners underscore the growing need for innovative pedagogical approaches such as DGBL in sales education. Sales roles are inherently performance-oriented and skills-based, requiring ongoing development to meet dynamic market demands (Cummins et al., 2013). DGBL offers a compelling response to several of these needs.

Engagement, Motivation, and Active Learning

Traditional training methods can fall short in sustaining high levels of learner engagement and motivation, both of which are pivotal for effective knowledge acquisition and skill development (Woo, 2014). DGBL, by leveraging interactive mechanics, narrative, challenge, and immediate feedback, can significantly boost active participation and foster intrinsic motivation (Tay et al., 2022; Chang et al., 2018). This heightened engagement is frequently proposed as a key factor leading to improved learning outcomes in various contexts, including corporate training (Hainey et al., 2016).

Safe and Realistic Practice for Complex Skill Development

Sales success often hinges on the mastery of interpersonal and strategic skills, such as negotiation, objection handling, and adapting communication to diverse client needs (Bussière, 2017). DGBL can offer simulations of realistic sales scenarios, providing a safe environment in which trainees can practice these complex skills without the real-world consequences of failure (Senderek et al., 2015; Dahalan et al., 2024). This sandboxed approach allows for experimentation, learning from mistakes, and iterative skill refinement, which is crucial for developing robust competencies (Chen et al., 2018).

Problem-Solving, Critical Thinking, and Learner Agency

Effective sales professionals are adept problem-solvers who can critically assess situations and devise appropriate strategies. DGBL can be designed to present learners with ill-defined problems or multifaceted challenges that require analytical thinking, strategic decision-making, and creative solutions (Noroozi et al., 2020). Such designs can also enhance learner agency by allowing individuals to make meaningful choices and observe the consequences of those choices within the game environment.

Scalability, Consistency, and Corporate Upskilling

Organizations often face substantial training demands as they seek to ensure that sales teams possess current knowledge and skills in response to new products, evolving market conditions, and the general need for continuous professional upskilling (Hernández and Moreno, 2019). DGBL, as a digital modality, offers a scalable and consistent way to deliver high-quality training content across geographically dispersed teams, thereby supporting efficient competency development (Tay et al., 2022).

The strategic integration of DGBL into sales education is therefore proposed not merely as a technological novelty to promote engagement and motivation, but as an informed approach to leverage its unique affordances in addressing organizational training needs, enhancing core sales competencies, and ultimately improving learning and performance outcomes in sales education.

Meta-analytic reviews offer qualified empirical support for these claims, though with important boundary conditions. Wouters et al. (2013) found that serious games outperformed conventional instruction on learning outcomes ($d = 0.29$) and retention ($d = 0.36$), with

benefits enhanced by multiple practice sessions, collaboration, and supplementary instruction. Merchant et al. (2014), in a meta-analysis of simulation-based learning, found an overall effect size of $d = 0.33$. However, both reviews focus predominantly on student populations in formal educational settings, and the Merchant analysis in particular draws heavily on classroom and undergraduate contexts. The applicability of these aggregate effects to professional adult learners in sales training—where learner agency, prior domain expertise, and motivation to engage are configured very differently—cannot be assumed without qualification. All et al.’s (2021) systematic review, which gives more attention to workplace and professional DGBL contexts, provides a more cautious aggregate picture and underscores the importance of design quality, instructional alignment, and methodological rigor in determining whether games outperform conventional alternatives. This study situates itself within that more critical tradition.

2.1.2 Current Approaches to Sales Education

Traditional and Experiential Methods

Sales education has matured well beyond rudimentary product training and now aims to develop the diverse competencies required in increasingly complex market environments. A variety of pedagogical strategies are employed to bridge theoretical knowledge with practical application, which is crucial for effective knowledge transfer from education to organizational performance (Tho and Trang, 2015).

Traditional methods such as lectures, assigned readings, and classroom discussions continue to play a role in delivering foundational knowledge, including core sales theories, ethical guidelines, and an understanding of market dynamics. However, the dynamic and action-oriented nature of sales warrants a strong focus on experiential learning. As reviewed by Cummins et al. (2013), sales pedagogy widely incorporates techniques intended to offer learners applied experience, such as role-play and simulation. These activities allow learners to practice sales interactions, from initial approaches to closing and objection handling, in a controlled environment, with the aim of building practical skill and often self-efficacy in handling sales situations (Knight et al., 2014).

Case Study Analysis

Analyzing real-world or hypothetical business cases helps learners develop diagnostic and strategic thinking skills that are directly relevant to sales challenges.

Field Projects and Internships

Direct engagement in sales activities, whether through internships or structured “selling lab” projects, provides invaluable hands-on experience and facilitates deeper understanding of the sales process through direct application (Bussière, 2017).

The emphasis on experiential methods reflects the view that active participation and “learning by doing” are critical for skill acquisition and retention in professional disciplines. These approaches are generally intended to foster not only procedural skills, but also critical thinking, problem-solving, and effective interpersonal communication.

Despite the recognized value of these methods, several challenges remain. Transfer from training environments to on-the-job performance can be inconsistent and is shaped by factors in both the training design and the organizational context (Tho and Trang, 2015). Ensuring the quality, consistency, and scalability of resource-intensive experiential methods such as deeply engaging role-play or mentored field experience can also be difficult for educational institutions and corporate training departments. Furthermore, assessing the skills developed through such methods often requires sophisticated rubrics and can remain partly subjective.

Adding to these pedagogical considerations is the ongoing evolution of the sales profession itself. The traditional linear “seven steps of selling” model (Moncrief and Marshall, 2005), while still useful, is increasingly supplemented or recontextualized by more dynamic, customer-centric, and digitally mediated approaches, particularly in complex B2B environments (Cardinali et al., 2025). This shift demands that sales education prepare learners for greater complexity, adaptability, and a deeper understanding of value co-creation.

It is in this context that DGBL presents itself as a promising pedagogical innovation. DGBL can encapsulate many principles of experiential learning within interactive and engaging formats. Moreover, it offers potential advantages in terms of scalability, consistency, immediate and individualized feedback, adaptive learning pathways, and the objective capture of rich performance data, thereby addressing some of the limitations of purely traditional methods.

2.2 Theoretical and Design Foundations

2.2.1 LM–GM Framework

In order to deliver the variety of skills and knowledge needed in sales education, a coherent design framework informed by learning theory is necessary in DGBL. Arnab et al. (2015) formulated the learning mechanics–game mechanics framework (LM–GM) by identifying a range of learning mechanics categorized by instructional purpose. The model shows how different game mechanics can be linked to corresponding learning mechanics across increasing levels of cognitive complexity.

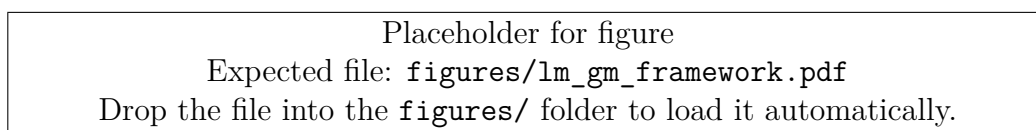


Figure 2.1: LM–GM model classified based on Bloom’s ordered thinking skills, adapted from Arnab et al. (2015).

Correspondingly, the framework outlines numerous game mechanics that can be employed to realize these learning mechanics. Examples include:

- **Guidance & Feedback:** Tutorials, hints, pop-up information, and direct feedback messages such as success/failure indicators or explanatory notes.
- **Participation & Action:** Quests, tasks, levels, puzzles, resource management challenges, and the use of tokens or point systems.
- **Exploration & Discovery:** Open environments, branching narratives, hidden information elements, and interactive objects.
- **Reflection & Collaboration:** In-game dialogue systems, forums, team-based challenges, or voting mechanisms.
- **Motivation & Ownership:** Customizable avatars, leaderboards, collection mechanics, and clear progression systems.

The LM–GM framework encourages a design process that begins with clearly defined learning outcomes, followed by the identification of the learning mechanics necessary to achieve

them. Appropriate game mechanics are then selected to implement these learning mechanics effectively. This helps ensure that the game’s design remains anchored to pedagogical purpose.

For instance, if a learning objective is for sales trainees to classify different types of customer objections, which involves learning mechanics such as identification and discrimination, the game might employ card-sorting challenges or scenario-based quizzes in which players must correctly categorize presented objections to progress. For objectives involving negotiation skills, which rely on experimentation and reflection, the game might use branching dialogue systems with varied outcomes and provide after-action reviews as a feedback mechanic to encourage reflection on choices and consequences.

Furthermore, the LM–GM framework serves not only as a design tool but also as an analytic tool for evaluating existing serious games. It allows researchers and practitioners to deconstruct how specific game features contribute, or fail to contribute, to learning, which is especially valuable when assessing the pedagogical soundness of DGBL interventions. In their own application of the framework to the serious game *Re-Mission*, Arnab et al. (2015) note that even within LM–GM-designed games, “extraneous (pedagogy-independent) game mechanics are often designed to enhance game play,” with the consequence that “learning occurs only tangentially.” This analytic capacity—identifying where game mechanics fail to activate their intended learning functions despite structural alignment—is one reason the framework is adopted as the theoretical lens for the intervention evaluated in this study, and why empirical testing is necessary to verify whether design-level alignment translates into cognitive-level learning gain.

2.2.2 Cognitive Load Theory

As described in the background of this study, cognitive load theory (CLT) is used as one of the main mechanisms that justifies the learning design based on DGBL, in the sense that the mediation of cognitive load through game design is expected to support higher self-efficacy and engagement. CLT begins with the understanding that learning, whatever mechanisms and conditions are used to produce it, depends on the reproduction and organization of knowledge in memory. Yet similar conditions can lead to different outcomes, which raises a cognitive-architectural question: how much working memory is required to learn something at a given time, and how does that relate to the limits of the human cognitive system?

Sweller et al. (2011) describe CLT on the basis that working memory is limited in both ca-

capacity and duration, whereas long-term memory is virtually unlimited. The crux of learning lies in transferring information from the temporary space of working memory into the more durable structures of long-term memory. This occurs through schema construction and, over time, through automation.

CLT categorizes cognitive load into three types: intrinsic, extraneous, and germane load. Intrinsic load is the inherent difficulty of the material, influenced by element interactivity. Extraneous load is created by poor instructional design, distracting materials, or unnecessary embellishment and should be minimized as much as possible. Germane load is the mental effort that directly contributes to learning and schema formation. Instruction should therefore aim to maximize this form of effort through strategies that promote reflection, deep processing, and transfer.

These distinctions enable educators and instructional designers to diagnose which type of load learners are experiencing, reduce extraneous load by streamlining content, scaffold intrinsic load to match learner readiness, and amplify germane load to support deeper learning. CLT thus becomes an invaluable tool not only for explaining how learning occurs, but for shaping how it can be facilitated effectively. While useful, CLT also has notable criticisms that should be acknowledged when designing learning materials. First, it depends on a relatively simple representation of memory and may underspecify emotional, episodic, or procedural influences on learning. Second, it risks circularity when failures are retrospectively labelled as extraneous load and successes as germane load. As Popper, discussed in Church (1975), argued more broadly, a theory that can explain everything risks becoming difficult to falsify. Despite these criticisms, CLT remains highly useful in designing instructional media, especially in fields such as sales education, where much knowledge is tacit and only partially formalized.

2.2.3 Self-Efficacy

Digital Game-Based Learning offers a promising avenue for fostering self-efficacy, particularly when it is designed in congruence with principles aimed at managing cognitive load so as to create an optimal environment for skill development and confidence building. As described by Bandura (1977), self-efficacy beliefs are primarily shaped through four sources of information: mastery experiences, vicarious experiences, verbal persuasion, and physiological or affective states.

DGBL interventions can be strategically designed to leverage these sources. For mastery

experiences, DGBL can provide repeated opportunities for successful task completion and problem-solving. By segmenting complex sales scenarios into manageable, scaffolded game tasks, learners can accumulate a sequence of scaffolded successes. These achievements build confidence in their own abilities, especially when tasks feel achievable because cognitive load has been managed appropriately.

DGBL can also facilitate vicarious experiences by incorporating models of successful performance, such as non-player characters or simulated peers effectively navigating difficult sales interactions. Similarly, verbal persuasion can be built into game feedback systems through encouraging messages, constructive guidance, and reinforcement for effective play. Finally, DGBL can help shape positive affective states by creating learning environments that are engaging and appropriately challenging without becoming overwhelming. By reducing cognitive overload and learning-related anxiety, and by supporting enjoyment or even flow-like experience (Chang et al., 2018), DGBL can help learners associate the learning process with positive feelings that support stronger efficacy beliefs.

This study therefore posits that a scaffolded DGBL approach, by carefully managing cognitive load and structuring opportunities for success, will provide more favorable conditions for mastery experiences and supportive affective states than traditional digital training methods, which in turn should support stronger self-reported self-efficacy.

2.2.4 Engagement

Alongside cognitive load and self-efficacy, learner engagement stands as a third critical factor influencing the effectiveness of educational interventions, particularly those employing digital and interactive modalities such as DGBL. Engagement in learning is broadly understood as the intensity and quality of a learner’s active involvement and commitment to a learning activity or process (Christenson et al., 2012). It is a multifaceted construct, often encompassing behavioral, emotional or affective, and cognitive dimensions (Fredricks et al., 2004), further described below:

- **Behavioral Engagement:** Observable actions such as participation in tasks, time on task, effort, persistence in the face of difficulty, and adherence to learning protocols.
- **Emotional/Affective Engagement:** Feelings and affective responses to the learning activity, including interest, enjoyment, enthusiasm, and a sense of connection to the task.

- **Cognitive Engagement:** Psychological investment in learning, including focused attention, use of learning strategies, self-regulation, and willingness to grapple with complex ideas and exert mental effort.

A substantial body of research indicates a strong positive relationship between learner engagement and desired educational outcomes, including achievement, skill acquisition, and knowledge retention (Hainey et al., 2016; Hernández and Moreno, 2019; Tay et al., 2022). In workplace training, fostering engagement is especially important because adult learners often adopt a pragmatic perspective and seek training that is relevant, stimulating, and clearly applicable to their professional roles (Knowles et al., 2020). Disengagement can lead to superficial learning, poor transfer, and a perception of wasted training investment.

DGBL is frequently cited for its potential to enhance learner engagement compared with more passive instructional methods (Dahalan et al., 2024; Liu and Zhou, 2025). Several characteristics of well-designed games contribute to this potential:

- **Intrinsic Motivation:** Games often tap into challenge, curiosity, fantasy, and control.
- **Clear Goals and Rules:** Games provide explicit objectives and rules that can improve focus and behavioral engagement.
- **Immediate Feedback:** Continuous feedback helps learners understand the consequences of their decisions and remain cognitively invested (Chang et al., 2020).
- **Sense of Agency and Control:** Games often let players make meaningful choices that affect progression or outcome.
- **Narrative and Storytelling:** Story-driven contexts can enhance emotional and cognitive engagement (Chen et al., 2018).
- **Challenge and Mastery:** Appropriately balanced challenge can motivate persistence and contribute to both affective and behavioral engagement.

This study therefore hypothesizes that the proposed DGBL intervention, through game mechanics aligned with learning objectives under the LM-GM framework, will foster higher levels of behavioral, emotional, and cognitive engagement than traditional click-and-continue digital training. This increased engagement is expected to contribute positively to learning outcomes and self-efficacy.

2.2.5 Sales Competency

The ultimate aim of most workplace training initiatives, including the DGBL intervention proposed in this research, is to bring about a positive change in learners' capabilities and subsequently in their job performance. In the domain of sales, this translates to the development and enhancement of sales competency. Competency can be understood as an integrated set of knowledge, skills, abilities, and related personal characteristics that enable effective task performance (Titus, 1994).

Sales competency is multifaceted. While specific emphases vary across roles and industries, several core areas recur in the literature (Cummins et al., 2013): product and market knowledge; sales process skills across prospecting, needs assessment, presentation, objection handling, negotiation, and closing; interpersonal and communication skills; problem-solving and strategic thinking; and self-management attributes such as resilience, proactivity, adaptability, and ethical conduct.

A key challenge in any training program is the valid and reliable assessment of competency development. Traditional assessments often focus on isolated knowledge recall, whereas a more holistic view of competency requires methods that can gauge the integrated application of knowledge and skill in realistic contexts. In DGBL, performance within game scenarios, choices made, strategies employed, and outcomes achieved can all provide meaningful indicators of targeted competency development (All et al., 2021).

The DGBL intervention designed for this study aims to impact several core sales competencies by simulating realistic scenarios and requiring active participation. In the finalized thesis, this broader competency domain is represented most directly through adaptive selling, transfer intention, and related applied sales outcomes reported in the results chapters.

2.3 Theoretical Integration

2.3.1 Cognitive Load and Self-Efficacy

Cognitive Load Theory and Self-Efficacy Theory, while originating from different psychological traditions, describe constructs that are deeply interdependent and reciprocally influential within learning environments, especially those involving complex skill acquisition such as corporate sales training.

The experience of cognitive load during learning can significantly affect self-efficacy. When learners are confronted with tasks that impose excessive extraneous cognitive load because of poorly structured information, confusing interfaces, or an overwhelming density of novel elements, their capacity to process information and engage in meaningful learning is hampered (Sweller et al., 2011). Such situations often lead to frustration, confusion, and a sense of incompetence. These negative learning experiences undermine mastery experiences, which Bandura (1977) identifies as the most powerful source of self-efficacy. Furthermore, excessive cognitive load can induce adverse physiological and affective states, such as anxiety or mental fatigue (Chang et al., 2018), which learners may interpret as evidence of inadequacy.

Conversely, existing self-efficacy can shape the perception and management of cognitive load. Learners with robust self-efficacy tend to approach challenging tasks with greater persistence, effort, and strategic thinking (Bandura, 1986). They may therefore allocate their cognitive resources more effectively and find difficult tasks engaging rather than debilitating. In contrast, learners with low self-efficacy may feel overwhelmed more readily and disengage sooner. Without effective cognitive load management, a downward spiral can occur: high load leads to frustration, which lowers self-efficacy, which in turn reduces persistence and increases the likelihood of further failure.

Instructional designs that actively manage cognitive load are therefore not merely improving information processing efficiency. They are also cultivating conditions more conducive to strong self-efficacy. By making learning tasks feel manageable and achievable, instructional interventions allow learners to experience incremental success and build a portfolio of mastery experiences, thereby supporting confidence and persistence.

The proposed DGBL intervention, with its emphasis on systematic scaffolding and phased increases in task complexity, is intended to interrupt negative cycles of overload and diminished confidence. Instead, it seeks to foster a positive spiral in which manageable cognitive demands support mastery experiences, positive affective states, stronger self-efficacy, greater engagement, and more effective application of learned skills.

2.3.2 DGBL as a Moderator of Cognitive Load

A key premise of using DGBL in complex training domains such as sales is its potential to strategically manage learner cognitive load, thereby creating a more effective and efficient learning experience than traditional methods that may be less adept at this regulation. Different genres of DGBL may offer distinct affordances in this regard; for instance, digital

card games can provide structured learning environments through their inherent mechanics (Kordaki and Gousiou, 2017).

Well-designed DGBL can minimize extraneous load by integrating information presentation directly into tasks, providing clear signposting, highlighting relevant interactive elements, and structuring progression clearly (Sweller et al., 2011). Digital card games, in particular, naturally present information in discrete units on the cards themselves, which can reduce cognitive clutter and focus attention. Early game levels or tutorial phases can also function as worked examples.

DGBL can also help manage intrinsic load through segmentation and sequencing as a form of scaffold. The act of drawing, holding, and playing cards one at a time in a digital card game inherently segments the decision-making process, allowing learners to focus on a limited set of options or information elements at any given point (Kordaki and Gousiou, 2017). This can be combined with a gradual introduction of more complex cards or rules so that the learner is not overwhelmed prematurely.

Finally, DGBL can foster germane load through active problem-solving and decision-making. Digital card games often require strategic thinking such as hand management, anticipating consequences, and planning combinations of actions, all of which can promote deeper cognitive processing. Repeated practice, immediate contextual feedback, and role-playing or narrative framing can further enhance emotional investment and meaningful understanding.

By strategically employing these design principles, DGBL can become not merely a motivational tool, but a powerful instructional environment designed to optimize the cognitive conditions for learning. The scaffolded card game in this research was intended to leverage these affordances to manage cognitive load more effectively than a standard click-and-continue format.

Chapter 3

Conceptual Framework and Hypothesis

This study is guided by a conceptual framework that posits the scaffolded Digital Game-Based Learning (DGBL) intervention, “Improve Your Sales Game,” will influence key learning outcomes by affecting intermediate psychological variables. Specifically, the DGBL intervention, through its theoretically grounded design emphasizing scaffolding, CLT principles, and LM–GM framework application, is expected to lead to lower perceived cognitive load and higher sales self-efficacy and learner engagement compared with a traditional digital training approach. These psychological variables are, in turn, hypothesized to be interrelated and to collectively contribute to the development of sales competency-related outcomes and a greater intention to apply learning.

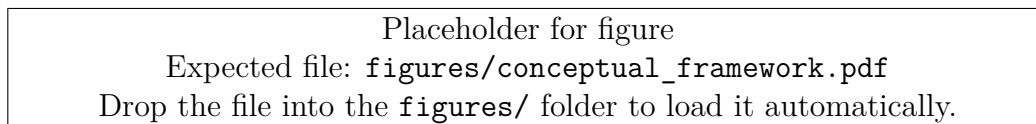


Figure 3.1: Conceptual framework of the DGBL intervention.

In the finalized thesis, the broad competency domain described in the original proposal is represented most directly through adaptive selling, transfer intention, and related applied sales outcomes.

Based on this framework and the research questions, the following hypotheses are proposed:

- **Hypothesis 1.** Participants in the scaffolded DGBL intervention (“Improve Your Sales Game”) will demonstrate more favorable sales training outcomes compared with

participants in the traditional click-and-continue digital training group. This hypothesis is supported if the scaffolded DGBL group reports significantly higher learner engagement, higher sales self-efficacy, lower perceived cognitive load, greater improvement in competency-related sales outcomes including adaptive selling, and a higher intention to apply learning.

- **Hypothesis 2.** Within the scaffolded DGBL sales-training condition, perceived cognitive load, learner engagement, and sales self-efficacy will be interrelated and will collectively contribute to competency-related development in theoretically consistent ways. This hypothesis is supported by significant intercorrelations in the expected directions between perceived cognitive load, sales self-efficacy, and learner engagement, and by positive associations of self-efficacy and engagement with adaptive selling, transfer intention, and related applied sales outcomes.

Chapter 4

Methodology

This chapter outlines the research methodology employed to address the research questions posed in the introduction of the study. It details the research design, the characteristics of the study participants and sampling strategy, the design of the experimental and control interventions, the instruments used for data collection, the experimental procedure, and the plan for data analysis.

4.1 Research Design

This study utilized a quantitative, experimental research design, specifically a pre-test, post-test randomized controlled trial. This design was selected for its recognized strength in establishing cause-and-effect relationships between an intervention and observed outcomes, and for its ability to minimize threats to internal validity (Kirk, 2013; Shadish et al., 2004).

The RCT involved two parallel groups:

- **Experimental Group:** Participants in this group received the scaffolded, game-based digital learning sales-training intervention, designed as a card-based digital game.
- **Control Group:** Participants in this group received a traditional click-and-continue digital training module covering the same broad sales concepts, but delivered in a non-gamified, linear format without the interactive scaffolding strategies or game mechanics of the experimental condition.

Participants were randomly assigned to either the experimental or control group. Data on key dependent variables, including self-efficacy, perceived cognitive load, learner engagement, transfer intention, adaptive selling, and other competency-related outcomes, were collected before the training intervention and immediately after its completion.

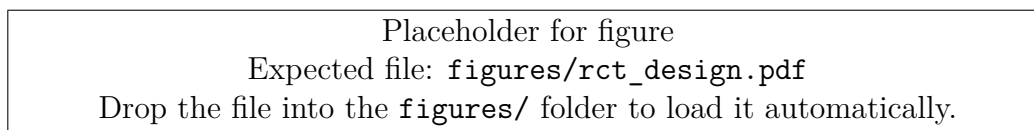


Figure 4.1: RCT design of DGBL in sales education.

The choice of an RCT design is justified by its capacity to address the study’s primary research question, which seeks to compare the effectiveness of the two training interventions. Random assignment helps ensure baseline equivalence on known and unknown confounding variables, thereby allowing more confident attribution of observed differences to the intervention itself. The study was pre-registered prior to data collection. Behavioral log data confirm that the first participant session was recorded on 31 March 2026 and the final session on 15 April 2026, establishing the data collection window as a two-week period in which all event-level records in the platform database were generated.

4.2 Population

The target population for this study comprised novice sales professionals or individuals aspiring to enter sales roles who had limited prior formal sales training or experience. Because the study was delivered online through a publicly accessible website, participation was open to any adult individual interested in sales training.

Participants were recruited through online channels, including professional networking sites, relevant online forums or communities focused on sales and professional development, and related digital recruitment routes. Eligibility for inclusion in the final analysis primarily depended on completion of the pre-test, the assigned training module, and the post-test measures.

No direct compensation was planned for participation, but participants benefited from free access to a sales training module. Participation was voluntary, and informed consent was obtained digitally before data collection began.

To characterize the sample, demographic information was collected as part of the pre-test

questionnaire. This included:

- Age group (e.g., 18–24, 25–34, 35–44, etc.)
- Years of experience in sales, if any (e.g., none, less than 1 year, 1–3 years, more than 3 years)
- Highest level of education achieved
- Familiarity with digital games for learning
- Current industry of work, if applicable

This demographic data was used descriptively to characterize the sample and to support secondary subgroup interpretation where feasible.

4.3 Interventions, Measures, and Procedure

4.3.1 Interventions

Participants were randomly assigned to one of two digital sales-training interventions: the experimental scaffolded DGBL condition or the control traditional digital training condition. Both interventions covered the same broad sales topics derived from established sales-process models, including prospecting, needs analysis, presentation of solutions, handling objections, and closing. The primary difference lay in instructional approach and delivery format.

The Scaffolded DGBL Intervention (Experimental Group)

The experimental intervention was a scaffolded, card-based digital game titled “Improve Your Sales Game.” The game was designed on the basis of Cognitive Load Theory to manage cognitive demands and Bandura’s Self-Efficacy Theory to foster learner confidence. The core of the game involved a deck of digital cards representing sales scenarios across industries and sales stages. The design process was guided by the LM–GM framework to ensure alignment between learning objectives and game mechanics.

The game unfolded in three distinct phases, each targeting different aspects of the sales process and associated skills:

Phase 1: Lead Collection and Retention (Conceptual Understanding & Identification)

- **Learning Objective:** To understand and identify different stages of a sales funnel and classify potential leads.
- **Gameplay:** Participants are presented with event cards describing customer scenarios or lead characteristics and must correctly classify those cards according to predefined sales stages to earn lead tokens. This phase emphasizes foundational knowledge and correct identification, with direct feedback on classification accuracy.

Phase 2: Sales Process Roleplay (Procedural Skill Application & Negotiation)

- **Learning Objective:** To apply sales techniques in simulated interactions, with a focus on negotiation and objection handling.
- **Gameplay:** Participants use collected lead tokens to engage with simulated customer scenarios. They draw action cards representing sales tactics, communication strategies, or solutions, and must strategically choose plays that successfully navigate the challenges posed by event cards. This phase is intended to provide scaffolded practice, immediate feedback, and mastery experiences.

Phase 3: Problem-Solving, Evaluation, and Reflection (Strategic Thinking & Metacognition)

- **Learning Objective:** To develop strategic thinking in response to complex or unexpected sales situations and reflect on optimal approaches.
- **Gameplay:** Participants may encounter crisis cards that introduce more complex dilemmas requiring broader strategic judgment. This phase encourages deeper processing, evaluation of alternative strategies, and reflection on the broader sales principles underlying the card-based interactions.

The overall design aimed to reduce extraneous cognitive load through clear rules and interface, manage intrinsic load by segmenting the sales process into phases and card-based interactions, and promote germane load through strategic decision-making and reflection, thereby fostering self-efficacy and engagement.

The Traditional Digital Training (Control Group)

Participants in the control group engaged with a digital training module that covered the same core sales topics as the DGBL intervention, but in a traditional click-and-continue linear presentation.

- **Format:** Information was presented through a series of text-based slides, with optional static images or diagrams, and learners progressed by clicking a “next” or “continue” button.
- **Content Delivery:** Concepts were explained directly. Practice opportunities, where included, took the form of simple quizzes rather than interactive scenario gameplay.
- **Lack of DGBL Elements:** This condition deliberately excluded the card mechanics, token systems, game phases, interactive challenges, and dynamic feedback structures present in the DGBL intervention.

This control condition represented a standard non-gamified digital learning approach, allowing for a clearer comparison of the added value of the scaffolded DGBL intervention.

4.3.2 Instruments and Measures

Data for this study were collected using questionnaires administered at pre-test and post-test, alongside behavioral and applied sales indicators. The original measurement design prioritized validated scales where available and theory-informed item development where needed.

- **Sales Self-Efficacy:** Participants’ confidence in performing key sales tasks was assessed with a 10-item scale adapted from Bandura’s self-efficacy guidelines and prior sales-education work (Bandura, 1977; Knight et al., 2014).
- **Perceived Cognitive Load:** Subjective cognitive load during the training intervention was assessed post-intervention using a 4-item scale informed by Paas et al. (2008), Chang et al. (2020), and Liu et al. (2023).
- **Learner Engagement:** This multifaceted construct was assessed post-intervention using a 6-item scale developed from Fredricks et al. (2004) and related DGBL research (Woo, 2014).

- **Sales Competency-Related Outcomes:** The original proposal specified a scenario-based competency test covering core sales skills; in the finalized thesis, the applied sales domain is represented most directly through adaptive selling and related post-intervention outcomes.
- **Transfer Intention:** Participants’ intention to apply learned sales skills in work contexts was assessed post-intervention using a 3-item scale adapted from Tho and Trang (2015).

The study combined these pre-test and post-test self-report instruments with behavioral log data from the training platform. All major scales showed acceptable to excellent internal consistency in the analytic sample.

Table 4.1: Measurement instruments and internal consistency

Scale	α	Items	Rating
Pre self-efficacy (1–5)	0.93	10	Excellent
Post self-efficacy (1–5)	0.96	10	Excellent
Post cognitive load (1–7)	0.84	4	Good
Post engagement (1–5)	0.92	6	Excellent
Post transfer intention (1–5)	0.95	3	Excellent
Pre ADAPTS-SV (1–7)	0.91	5	Excellent
Post ADAPTS-SV (1–7)	0.92	5	Excellent

Self-efficacy was measured with a 10-item sales self-efficacy scale administered at pre-test and post-test on a 5-point agreement scale, drawing on Bandura’s self-efficacy guidelines and prior sales-education work (Bandura, 1977; Knight et al., 2014). **Adaptive selling** was measured using ADAPTS-SV, a validated 5-item psychometric scale assessing adaptive selling beliefs on a 7-point agreement scale (Spiro and Weitz, 1990; Robinson et al., 2002), administered at both pre-test and post-test. The short-form version was selected deliberately: its brevity minimises additional survey burden and reduces the risk of attrition from survey fatigue, which is a particular concern in voluntary online training contexts. **Cognitive load** was measured post-test using four items assessing mental effort, perceived difficulty, complexity, and mental strain, following standard subjective load measurement logic (Paas et al., 2003; Paas et al., 2008; Sweller et al., 2011). **Engagement** was assessed post-test using six items spanning behavioral, emotional, and cognitive engagement, consistent with multidimensional engagement theory (Fredricks et al., 2004; Woo, 2014). **Transfer intention** was assessed post-test using three items measuring the intention to apply learned

sales skills in future practice, adapted from transfer-focused workplace learning research (Tho and Trang, 2015).

The survey instrument file indicates a 7-point format for cognitive load and a 5-point format for transfer intention, and these codings are consistent with the finalized analysis report. This manuscript therefore follows the finalized report as the authoritative source for scale interpretation and scoring.

4.3.3 Procedure

The study was conducted entirely online. The following steps outline the procedure for participant involvement.

Recruitment and Initial Contact

- Participants were recruited through online channels such as professional networking sites, online forums, and related digital communities.
- Recruitment materials provided a brief overview of the study’s purpose, the voluntary nature of participation, the estimated time commitment, and the benefit of free access to a sales training module.
- A link to the study website was provided in the recruitment materials.

Informed Consent and Eligibility

- Upon accessing the study website, potential participants were presented with an information sheet explaining the research, data usage, confidentiality, withdrawal rights, and researcher contact details.
- Participants then provided digital informed consent before proceeding.
- A brief screening question was used to ensure that participants were adults.

Pre-Test Data Collection

- After providing consent, participants completed an online pre-test questionnaire.

- The pre-test included demographic information, baseline self-efficacy, and baseline applied sales measures.

Random Assignment

Upon successful completion of the pre-test, participants were randomly assigned to one of the two intervention conditions:

- Experimental Group: Scaffolded DGBL intervention (“Improve Your Sales Game”).
- Control Group: Traditional click-and-continue digital training.

The randomization was managed automatically by the online platform hosting the study using a simple assignment algorithm based on time and sequence of access.

Training Intervention Phase

Participants were then immediately directed to their assigned digital training module.

- The experimental group played the designed DGBL “Improve Your Sales Game.”
- The control group completed the click-and-continue sales training module.

Both interventions were designed to cover the same broad sales content and were intended to be completed in a single session.

Post-Test Data Collection

Immediately after completing the assigned intervention, participants completed an online post-test questionnaire. This included post-training self-efficacy, cognitive load, engagement, transfer intention, and competency-related or adaptive sales measures.

Participant Flow and Analytic Sample

Participant flow is summarized in a CONSORT-style format to clarify recruitment, randomization, attrition, and the final analytic sample.

The platform recorded 179 individuals who accessed the study website. Of these, 124 completed the demographic form and 92 completed the pre-test and were randomized. Allocation favored the game condition numerically because of the observed platform flow during recruitment: 55 participants were assigned to the game condition and 37 to the control condition. Non-completion was substantially higher in the game condition. Twenty-nine of the 55 participants allocated to the game condition did not provide usable post-test data, compared with 8 of the 37 in the control condition. The final per-protocol analytic sample therefore comprised 63 participants: 32 in the game group and 31 in the control group.

This attrition pattern is substantively important. The game group experienced significantly higher attrition than the control group (52.7% vs. 21.6%; Fisher’s exact test, odds ratio = 0.28, $p = .010$). In the game arm, dropout was concentrated during phase 2 of the matching game (41% of game dropouts), phase 1 card classification (24%), and cases in which participants did not engage meaningfully with the game at all (21%). In the control arm, most dropouts occurred during slide training (63% of control dropouts), followed by attrition after the pre-test (25%).

Baseline self-efficacy did not differ significantly between completers and non-completers, suggesting that attrition was not obviously ability-selective. Nevertheless, the differential attrition implies that the final game sample may represent a more persistent or motivated subset of those initially randomized. Accordingly, the primary analyses reported in this thesis are per-protocol analyses based on completers, with the non-completion pattern interpreted as an outcome-relevant feature of the intervention rather than merely a procedural inconvenience.

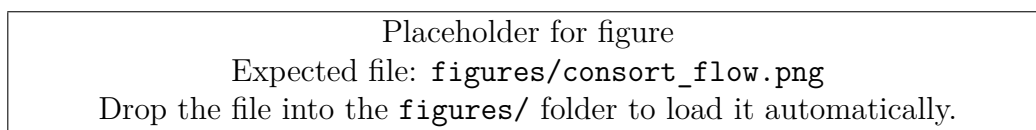


Figure 4.2: CONSORT-style participant flow diagram.

4.4 Data Analysis Plan

Prior to hypothesis testing, data were screened for accuracy, missing values, and the assumptions of statistical tests. Descriptive statistics were calculated for demographic variables and key outcome measures at pre-test and post-test for both groups. An alpha level of .05 was used for tests of statistical significance.

Table 4.2: Hypothesis and Analysis Methods

Focus	Variable(s)	Statistical analysis
RQ1 / H1 between-group comparisons	Self-efficacy, cognitive load, engagement, transfer intention, and competency- related outcomes	Between-group comparisons of post-test outcomes and pre-post change scores across conditions.
RQ2 / H2 interrelationships within the DGBL group	Perceived cognitive load, learner engagement, sales self-efficacy, and competency- related outcomes	Pearson correlation coefficients to examine the strength and direction of relationships.
Collective contribution of psychological variables	Cognitive load, engagement, self-efficacy predicting competency- related outcomes	Multiple regression analysis to assess collective and unique contributions.
Channeling effect exploration	Cognitive load, self-efficacy, engagement, competency- related outcomes	Exploratory follow-up interpretation based on observed correlation patterns.

The finalized analysis report treated the final completer sample as the primary analytic set ($N = 63$). For the two primary pre-post outcomes—sales self-efficacy (SE) and adaptive selling (ADAPTS)—between-group comparisons used ANCOVA with the respective pre-test score as a covariate, following CONSORT recommendations for RCTs that include baseline measurement. ANCOVA was preferred over gain-score t -tests because it does not impose the implicit constraint that the post-on-pre regression slope equals 1.0, and because partitioning pre-test variance from the error term increases statistical precision even when baseline groups are well-balanced. For the post-test only measures—cognitive load, engagement, and transfer intention—Welch’s independent samples t -tests were used, as no pre-test baseline was available for these constructs. Within-group pre-post changes on self-efficacy and adaptive selling were tested using paired t -tests.

Exploratory analyses—including cognitive load item decomposition, correlation moderation via Fisher z tests, k -means cluster-based expertise profiling ($k = 3$), and bootstrapped mediation analysis within the game group—were conducted as theoretically motivated follow-ups to the primary hypotheses. These analyses were not pre-registered and are reported as exploratory throughout. Sensitivity checks excluded four participants with extreme completion times ($>10,000$ seconds, likely left the browser open); this produced an $n = 59$ sample whose results were consistent in direction and comparable in magnitude to the primary analysis.

Chapter 5

Results

5.1 Baseline Equivalence

The two analyzed groups were statistically comparable at baseline. Welch’s tests showed no significant differences in pre-test self-efficacy, pre-test adaptive selling, or recorded time on task prior to the outcome analyses. This supports the integrity of randomization among completers, while not eliminating the interpretive importance of differential post-randomization attrition.

Table 5.1: Baseline equivalence between analyzed groups (0–100 scale)

Measure	Game M (SD)	Control M (SD)	t	p
Pre self-efficacy (%)	64.1 (20.0)	62.6 (17.9)	0.33	.746
Pre ADAPTS score (%)	63.1 (20.4)	62.2 (18.4)	0.20	.851
Time on task (sec)	3592 (9624)	3655 (12333)	-0.02	.982

Note. Self-efficacy and ADAPTS pre-test scores are reported on a 0–100 scale for consistency with the ANCOVA analyses in §5.2. Both groups are well-matched at baseline. Because ANCOVA is used regardless of balance, this check serves as descriptive confirmation rather than an analytical gate.

5.2 Hypothesis 1: Between-Group Comparisons

Hypothesis 1 predicted that the scaffolded game would produce more favorable post-test outcomes than the conventional digital training format.

Primary Outcomes: ANCOVA (SE and ADAPTS)

For the two primary pre-post outcomes, ANCOVA with pre-test as covariate provided the primary test. Table 5.2 shows results with both unadjusted descriptives and ANCOVA-adjusted means.

Table 5.2: ANCOVA results for primary outcomes (0–100 scale; $n = 63$)

Outcome	Game M	Ctrl M	Adj. Game	Adj. Ctrl	$F(1,60)$	p	η_p^2
SE Post (%)	70.2 (17.5)	75.0 (13.5)	69.9	75.5	3.56	.064†	.056
ADAPTS Post (%)	67.9 (19.3)	72.3 (13.7)	67.7	72.5	2.22	.141	.036

Note. Adjusted means estimated at grand mean pre-test score (SE: 63.4; ADAPTS: 62.7). Pre-test covariate $\beta = 0.56$ for both outcomes; model $R^2 = .42-.46$. † $p < .10$.

SE showed a marginal trend favouring the control condition ($F(1,60) = 3.56$, $p = .064$ †, $\eta_p^2 = .056$), approaching a medium effect size. ADAPTS did not differ significantly between groups ($F(1,60) = 2.22$, $p = .141$, $\eta_p^2 = .036$, small effect). These results do not support H1 for the full sample, but the SE effect is not trivially small: at $\eta_p^2 = .056$, the study is underpowered to reliably detect or rule out a meaningful group difference. A replication targeting 80% power at $\alpha = .05$ would require approximately $n = 120$.

Sensitivity Analysis: Intention-to-Treat Estimation

The per-protocol analysis above is restricted to the $N = 63$ participants who completed both the pre-survey and the full training session. To assess robustness to differential attrition (52.7% game vs. 21.6% control), two intention-to-treat (ITT) sensitivity analyses were conducted.

ITT-A: no-gain imputation ($n = 80$). Seventeen non-completers (15 game, 2 control) submitted a pre-survey before discontinuing; their post-test scores were not collected. These

participants were included under a conservative no-gain assumption: post-test imputed equal to pre-test (i.e., the training produced zero change for that individual). The remaining 21 non-completers who left before completing the pre-survey were excluded from ITT-A, as no baseline data existed to anchor an imputation.

ITT-B: worst-case bounding ($n = 101$). All 38 non-completers were included under extreme assumptions: game non-completers received the lowest observed post-test score on each outcome, control non-completers received the highest. This scenario is an upper-bound exercise — it represents the most pessimistic possible estimate for the game condition — rather than a realistic estimate of the true treatment effect.

Table 5.3 reports ANCOVA results for SE and ADAPTS under all three scenarios.

Table 5.3: ITT sensitivity analysis: ANCOVA under three attrition scenarios

Scenario	N	Adj. Δ (G–C)	$F(1,df)$	p	η_p^2
<i>Self-efficacy (post SE $\sim$ condition + pre SE)</i>					
PP: completers only	63	–5.6%	$F(1,61) =$ 3.56	.063†	.056
ITT-A: no-gain ($n = 80$)	80	–5.9%	$F(1,78) =$ 4.69	.035*	.057
ITT-B: worst-case ($n = 101$)	101	–20.5%	$F(1,99) =$ 31.94	<.001***	.244
<i>Adaptive selling (post ADAPTS $\sim$ condition + pre ADAPTS)</i>					
PP: completers only	63	–4.9%	$F(1,61) =$ 2.22	.139	.036
ITT-A: no-gain ($n = 80$)	80	–2.8%	$F(1,78) =$ 0.91	.344	.012
ITT-B: worst-case ($n = 101$)	101	–20.6%	$F(1,99) =$ 24.77	<.001***	.200

Note. Adj. Δ = adjusted mean difference (game minus control) in percentage-point terms at grand mean pre-test covariate. PP = per-protocol. ITT-A imputes post = pre for 17 partial completers (pre-survey submitted, training not finished). ITT-B places all 38 non-completers at worst/best observed outcomes for game/control respectively. † $p < .10$, * $p < .05$, *** $p < .001$.

The ITT-A analysis strengthens the SE finding: the marginal per-protocol trend ($p = .063$ †) becomes significant ($p = .035$ *) once the 17 partial completers are included under no-gain imputation. The adjusted mean difference in SE is virtually unchanged (–5.6% vs. –5.9%), indicating that the marginal effect in the completer sample was attenuated by a positively

selected completer pool rather than inflated by it. The ADAPTS null finding remains non-significant under ITT-A ($p = .344$), confirming that the per-protocol result is not an artefact of survivor selection. Under ITT-B, both outcomes reach significance, but the extreme effect sizes ($\eta_p^2 = .24$ for SE, $.20$ for ADAPTS) reflect the mechanical consequence of assigning all no-data non-completers to floor and ceiling values and should not be interpreted as realistic estimates. Taken together, the ITT analyses confirm that the per-protocol findings are conservative: if anything, the true SE disadvantage for the game condition is slightly larger than the completer-only analysis suggests.

Secondary Outcomes: Welch t -Tests

Cognitive load, engagement, and transfer intention (post-test only) were compared using Welch’s t -tests (Table 5.4).

Table 5.4: Between-group post-test comparisons: secondary outcomes

Outcome	Game M (SD)	Control M (SD)	t	p	d
Post cognitive load (1–7)	4.81 (0.83)	4.19 (0.93)	2.83	.006**	0.71
Post engagement (1–5)	3.79 (0.61)	4.03 (0.61)	-1.54	.128	-0.39
Post transfer intention (1–5)	3.80 (0.69)	4.01 (0.62)	-1.27	.209	-0.32

The only statistically significant between-group difference was cognitive load: the game group reported substantially higher perceived cognitive demand ($t(59.8) = 2.83$, $p = .006$, $d = 0.71$) than the control group—a result opposite to the hypothesized direction. Engagement and transfer intention did not differ significantly, though both favoured the control condition with small-to-medium effects.

Cognitive Load Decomposition (Exploratory)

Item-total correlation analysis identified CL item 1 (mental effort: “How much mental effort did you invest?”) as partially distinct from items CL2–CL4, which capture perceived difficulty, complexity, and strain. A post-hoc exploratory decomposition separated an *intrinsic load* subscale (CL2 + CL3, $\alpha = .81$) from a *effort/strain* indicator (CL1 + CL4, $\alpha =$

.31). Between-group comparisons on each revealed a dissociation: the intrinsic load subscale differed significantly and substantially ($t = 3.52$, $p = .0004$, $d = 0.90$), whereas the effort indicator (CL1 alone, mental effort) did not ($p = .638$, $d = 0.13$). This pattern—equivalent directed effort, yet higher perceived difficulty and complexity—is consistent with extraneous load imposed by the game’s unfamiliar mechanics rather than added germane effort. This analysis was not pre-registered and should be treated as exploratory.

5.3 Within-Group Change

Both groups improved significantly from pre-test to post-test on self-efficacy and adaptive selling, confirming that both interventions were educationally active. The control group showed larger absolute gains on both outcomes.

Table 5.5: Within-group pre-post improvement summary (0–100 scale)

Group	Measure	Pre M (SD)	Post M (SD)	Gain
Game ($n = 32$)	Self-efficacy (%)	64.1 (20.0)	70.2 (17.5)	+6.2**
Game ($n = 32$)	ADAPTS score (%)	63.1 (20.4)	67.9 (19.3)	+4.8
Control ($n = 31$)	Self-efficacy (%)	62.6 (17.9)	75.0 (13.5)	+12.5***
Control ($n = 31$)	ADAPTS score (%)	62.2 (18.4)	72.3 (13.7)	+10.1

Note. Within-group SE change: game $p = .003^{**}$; control $p < .001^{***}$. Standardized within-group gains: SE game $d = 0.52$, control $d = 0.74$; ADAPTS game $d = 0.42$, control $d = 0.54$.

Both interventions produced meaningful pre-post improvement, but the control group’s gains were approximately twice as large on self-efficacy. This pattern is consistent with the ANCOVA adjusted means (69.9 vs. 75.5 for SE), where the marginal between-group difference ($p = .064^{\dagger}$) reflects the control group’s larger gain rather than a starting-point advantage. The aggregate picture therefore is not that the game harmed learning, but that it was less efficient for this participant mix — a distinction elaborated by the expertise-conditional analysis in Section 5.6.

5.4 Hypothesis 2: Interrelationships Within the Game Group

Hypothesis 2 proposed that cognitive load, engagement, and self-efficacy would relate to one another in theoretically expected ways within the game condition. The results partially supported this hypothesis. The positive relationships among self-efficacy, engagement, transfer intention, and adaptive selling were strong and coherent. However, cognitive load did not show the expected negative relationships with self-efficacy or engagement.

Table 5.6: Selected correlations within the game group

Variable pair	<i>r</i>	<i>p</i>	Interpretation
Cognitive load ↔ self-efficacy	.043	.814	Not significant; opposite to expected direction
Cognitive load ↔ engagement	.236	.184	Not significant; opposite to expected direction
Self-efficacy ↔ engagement	.761	< .001	Strong positive association
Self-efficacy ↔ transfer intention	.918	< .001	Very strong positive association
Engagement ↔ transfer intention	.833	< .001	Strong positive association
Self-efficacy ↔ ADAPTS post-test	.877	< .001	Very strong convergent validity
Engagement ↔ ADAPTS post-test	.736	< .001	Strong positive association
Cognitive load ↔ ADAPTS post-test	.172	.340	Not significant

The strongest message from these correlations is that the motivational and attitudinal constructs formed a tightly connected network inside the game condition. Learners who felt more capable also tended to feel more engaged, more willing to transfer their learning, and more adaptive in their perceived selling approach. What was not supported was the assumption that lower cognitive load would accompany these benefits. This weakens the argument that the game's scaffolding successfully managed mental effort.

Condition Moderation of Construct Relationships (Fisher z)

Fisher z tests compared key bivariate correlations between the game and control groups to assess whether the game restructured the psychological architecture of learning (Table 5.7).

Table 5.7: Condition moderation of construct relationships (Fisher z tests)

Relationship	Game r	Ctrl r	Fisher z	p	Sig.
SE $\times$ ADAPTS	.877***	.532**	2.90	.004	**
ENG $\times$ ADAPTS	.736***	.305†	2.37	.018	*
CL $\times$ ENG	.236 ns	.592***	-1.66	.097	†

The game condition significantly strengthened the coupling between self-efficacy and adaptive selling ($r = .877$ vs. $.532$, $p = .004$): confidence and adaptive selling beliefs were more tightly integrated in game completers than in control participants. Engagement–ADAPTS coupling was similarly amplified ($p = .018$). Conversely, the strong cognitive load–engagement link present in the control condition ($r = .592$) was attenuated in the game group ($r = .236$, $p = .097$ †), suggesting that game mechanics partially decouple engagement from cognitive effort, consistent with self-determination theory: the game generates intrinsic engagement through its own mechanics rather than through effortful processing.

Mediation: SE as the Mechanism Linking Engagement to Adaptive Selling

Bootstrapped mediation analysis ($B = 5,000$) within the game group tested whether SE mediates the effect of engagement on adaptive selling. The indirect effect (ENG $\rightarrow$ SE $\rightarrow$ ADAPTS) was significant (indirect effect $ab = 1.10$, 95% CI [.43, 2.24]). When SE was entered as a mediator, the direct effect of engagement on ADAPTS dropped from $\beta = 1.41$ to $\beta = 0.31$ (ns), indicating full mediation. SE alone explained $R^2 = .77$ of adaptive selling variance ($\beta = .79$, $p < .001$). No cognitive load mediation pathways were significant. This result should be treated as exploratory given $n = 32$ and wide confidence intervals, but the direction is theoretically coherent: the game’s primary within-group learning mechanism runs through confidence-building rather than cognitive load management.

5.5 Behavioral Evidence

The behavioral logs provide context for interpreting why the game may not have produced superior outcomes. In phase 1, average classification accuracy was 57.9%, only modestly above a 50% chance baseline, indicating that the task was genuinely challenging. Performance varied substantially by sales stage, with follow-up and needs analysis performing relatively well but presentation and pre-approach performing poorly. Faster responders tended to be more accurate and reported higher post-test self-efficacy; the correlation between mean phase-1 response time and post-test self-efficacy was $r = -.37$, $p = .031$. This pattern supports the interpretation that slower, more effortful gameplay was linked to lower confidence.

In phase 2, scenario performance also varied by encounter type. Some event types produced near-perfect win rates, whereas others were much more difficult. Conversion success and win rate correlated positively with engagement, suggesting that behavioral success aligned with a more positive learning experience. However, overall gameplay performance did not consistently predict the main learning outcomes strongly enough to treat in-game success as a reliable proxy for learning.

The control module warrants a more careful characterisation than a passive “click-through” format suggests. The module was designed with genuine learner agency: participants could navigate content non-linearly via a menu structure, access slides in any order, skip sections they judged to be already understood, and re-visit earlier material. Phase 2 of the control condition consisted of scenario-based decision prompts that directly paralleled the game’s Phase 2 encounter structure, applying the same sales content in a branching-choice format. The control was therefore a scaffolded self-directed module with scenario elements, not a purely passive didactic sequence.

Behavioral logs reflect this design. All 31 analyzed control participants viewed all 16 slides; inter-slide timing varied considerably. Twelve participants (39%) progressed with a mean inter-slide gap below 10 seconds, consistent with skipping material they had prior familiarity with or navigating rapidly before returning, while only six participants (19%) averaged more than 30 seconds per slide. This heterogeneity in pacing is consistent with agentic navigation rather than uniform disengagement. The control group’s equal or better average outcomes in this context therefore reflect a design that offered accessible, self-paced content delivery with genuine interactivity, providing a meaningful comparison to the game rather than a minimal-effort baseline.

5.6 Expertise-Conditional Effects

The aggregate between-group comparisons in Section 5.2 obscure substantial heterogeneity across learner expertise levels. Two complementary analyses address this: a continuous moderation analysis and an exploratory cluster analysis. The continuous analysis is treated as primary because it makes no discretisation assumptions and is not hostage to cluster-seed choices.

Continuous Moderation Analysis (Primary)

OLS regression models of the form $post_outcome \sim intercept + condition + pre_outcome_{centred} + condition \times pre_outcome_{centred}$ were estimated for each primary outcome. A significant interaction term indicates that the game-versus-control difference depends linearly on pre-intervention expertise level — the defining test for expertise moderation.

Both primary outcomes yielded significant interactions. For adaptive selling (ADAPTS), the condition $\times$ pre-ADAPTS interaction was $F(1, 59) = 11.65, p = .0012^{**}$ ($\hat{\beta}_{int} = 0.541$). For self-efficacy (SE), the condition $\times$ pre-SE interaction was $F(1, 59) = 5.52, p = .022^*$ ($\hat{\beta}_{int} = 0.365$). Simple slopes derived from each model characterise the direction and magnitude of the expertise-conditional pattern (Table 5.8).

Table 5.8: Simple slopes: game vs. control advantage at three pre-intervention expertise levels

Outcome	Low (−1 SD)	Mean (0 SD)	High (+1 SD)
ADAPTS (pp)	−15.17	−4.83	+5.51
SE (%)	−0.497	−0.224	+0.049

Note. Slope values are the estimated game-minus-control difference in outcome units at each level of pre-intervention expertise. Negative values indicate control advantage; positive values indicate game advantage. ADAPTS is expressed in percentage points; SE in standardised units.

The pattern is monotone for both outcomes: the game disadvantages low-prior-knowledge participants, produces near-equivalent outcomes at the mean, and trends positively for high-prior-knowledge participants. This continuous evidence corroborates the cluster-based findings below without relying on arbitrary group boundaries.

Exploratory Cluster Analysis (Supporting)

An exploratory k -means cluster analysis ($k = 3$, silhouette = 0.49) was conducted on two z -standardized *pre-intervention* features only: Pre-SE and Pre-ADAPTS. Restricting clustering to pre-intervention features avoids circularity: including post-test scores or gain variables in the cluster solution would partially define clusters by the very outcomes used to evaluate them. The $k = 3$ solution was selected via elbow and silhouette criteria; $k = 4$ isolated a single outlier rather than a theoretically meaningful fourth subgroup. Silhouette = 0.49 indicates acceptable cluster separation on baseline expertise dimensions. Cluster stability was formally verified by re-running k -means 100 times with different random seeds: the adjusted Rand index (ARI) across all seed pairs was 1.000 and per-participant label agreement was 100%, confirming that the solution is perfectly stable and not an artefact of initialisation.

Table 5.9: Cluster profiles: expertise-stratified learner groups ($k = 3$, $n = 63$)

Cluster	n	G/C	Pre-SE	Pre-ADAPTS	Profile
C1 (Novice)	10	4G / 6C	2.31 (low)	33% (low)	Novice Learners
C2 (Moderate)	28	15G / 13C	3.40	57%	Mid-level Learners
C3 (Advanced)	25	13G / 12C	4.17 (high)	81% (high)	Experienced Experts

Within-stratum game vs. control comparisons on ADAPTS gain revealed a monotone reversal across expertise levels, consistent with expertise reversal theory (Kalyuga et al., 2003):

Table 5.10: Within-stratum game vs. control: ADAPTS gain by expertise cluster (small-sample corrected effect sizes)

Expertise stratum	Game gain	Control gain	Hedges' g	95% CI	p
Low — C1 (Novice, $n = 10$)	+6.7%	+33.9%	−1.70	[−3.17, −0.23]	.001
Moderate — C2 ($n = 28$)	+7.6%	+9.7%	−0.15	[−0.89, +0.60]	n.s.
High — C3 (Advanced, $n = 25$)	−1.4%	−1.0%	+0.23	[−0.56, +1.02]	n.s.

Note. Hedges' g applies Hedges' (1981) small-sample correction to Cohen's d , dividing by $J = 1 - 3/(4df - 1)$. 95% CIs computed via the Hedges–Olkin variance formula. C1 raw $d = -1.88$; correction factor $J = 0.903$ ($df = 8$).

Novice learners in the game condition gained substantially less on adaptive selling than their control counterparts ($g = -1.70$, 95% CI $[-3.17, -0.23]$, $p < .001$), a large effect consistent with expertise reversal predictions. The wide confidence interval reflects the small C1 stratum ($n = 10$; $n_{\text{game}} = 4$), and the true effect may be larger due to survivor bias in the game novice group. The game also disadvantaged novices on engagement ($g = -1.86$, 95% CI $[-3.37, -0.36]$, $p = .001$) and transfer intention ($g = -1.32$, 95% CI $[-2.71, +0.07]$, $p = .022$). Moderate learners showed no meaningful group difference ($g = -0.15$, CI spanning zero). Advanced learners showed a small, non-significant edge for the game condition on ADAPTS ($g = +0.23$, CI spanning zero).

The proposed mechanism is consistent with CLT: for novice learners (C1), the game’s card mechanics, token economy, and phase structure impose extraneous load before domain schemas exist, overwhelming working memory, producing frustration rather than mastery experiences, and ultimately blocking learning. For advanced learners (C3), existing schemas absorb the same mechanics with minimal extraneous overhead, making the interactive design useful rather than harmful.

These cluster findings should be treated as exploratory for three reasons. First, the cluster solution was not pre-registered. Second, silhouette = 0.49 indicates acceptable but not strong cluster separation; however, stability across 100 random seeds was formally verified and found to be perfect: the adjusted Rand index (ARI) was 1.000 across all seed pairs and per-participant label agreement was 100%, confirming that the solution reflects genuine structure in the data rather than initialisation artefacts. Third, a survivor-bias caveat applies to the novice game cluster: game-group novices who remained in the analysis ($n = 4$) are the most persistent subset of those randomised, so the true disadvantage for novice game participants may be larger than observed. Conducting an ITT-based stratified analysis is recommended before making strong expertise-reversal claims.

To assess construct validity of the cluster solution, the three clusters were cross-tabulated against three demographic variables not used in the clustering. Sales experience showed a significant, monotone association with cluster membership ($\chi^2(6) = 16.83$, $p = .001$), with mean sales experience increasing from C1 (novice) to C3 (advanced) — confirming that the pre-intervention expertise clusters map onto an independently measured indicator of relevant prior knowledge. In contrast, game familiarity ($\chi^2(6) = 1.83$, $p = .886$) and age group ($\chi^2(6) = 9.41$, $p = .162$) showed no significant association with cluster membership, indicating that the clusters are not confounded by gaming experience or age demographics.

Chapter 6

Discussion

6.1 Summary of Findings

This thesis set out to test two linked propositions: first, whether a scaffolded digital game-based intervention would outperform a conventional digital training format in B2B sales education; and second, whether the internal pattern of psychological outcomes inside the game condition would reflect the theoretical framework derived from cognitive load theory, self-efficacy theory, engagement research, and LM-GM design logic. Both hypotheses require a nuanced verdict.

H1 was not supported at the aggregate level. ANCOVA revealed a marginal trend favouring the control group on self-efficacy ($F(1,60) = 3.56, p = .064^\dagger, \eta_p^2 = .056$), with no significant difference on adaptive selling. The game produced significantly higher cognitive load than the control ($d = 0.71$) without corresponding gains in engagement or transfer intention. However, aggregate comparisons mask expertise-conditional effects. Exploratory cluster analysis revealed that the null-to-marginal aggregate finding conceals a strong disadvantage for novice learners ($d = -1.88$ on ADAPTS gain) alongside neutral-to-positive trends for intermediate and advanced learners. The core story is therefore not that the game failed unconditionally, but that its effectiveness is contingent on learner expertise.

H2 received partial support. Within the game group, self-efficacy, engagement, transfer intention, and adaptive selling formed a tightly coherent positive network, corroborated by the finding that SE fully mediated engagement effects on adaptive selling (indirect effect = 1.10, 95% CI [.43, 2.24]). Cognitive load did not show the expected negative association with other outcomes, suggesting that subjective load captured a more complex experience

than uniform overload. Fisher z moderation tests further showed that the game significantly tightened the SE–ADAPTS coupling ($r = .877$ vs. $.532$, $p = .004$), indicating that the game restructured the psychological architecture of learning even when aggregate means were comparable.

The high attrition in the game condition (41.8% vs. 16.2% in control) is not merely a methodological footnote; it is part of the substantive evaluation. A format that loses a substantial share of participants before completion—disproportionately novices, disproportionately in phase 2—presents real barriers to organizational deployment, irrespective of completer outcomes.

6.2 Interpretation Through Cognitive Load Theory

The clearest between-group result was higher cognitive load in the game condition, a finding directly opposed to the intervention’s design intent. The exploratory CL decomposition adds important specificity: the game elevated *perceived difficulty and complexity* (CL2+CL3 intrinsic subscale, $d = 0.90$, $p < .001$) substantially more than *directed mental effort* (CL1, $d = 0.13$, ns). In CLT terms, this pattern is the signature of *extraneous* rather than *germane* load: participants found the game harder and more complex, but did not invest more deliberate effort than control learners. The mechanics appear to have added interface burden without adding productive cognitive engagement. Although this decomposition is exploratory and the effort subscale had poor internal consistency ($\alpha = .31$), the dissociation is theoretically coherent and points to a specific design target for future revisions.

The behavioral data reinforce this reading. Slower processing in the game was associated with lower post-test self-efficacy ($r = -.37$, $p = .031$), and phase-1 classification accuracy averaged only 57.9%, only modestly above chance. Many learners appear to have been still mastering the task structure while they were meant to be engaging with sales content. For novice learners specifically, the game concentrated too much difficulty into the early stages before domain schemas existed to absorb it—a pattern consistent with expertise reversal predictions (Kalyuga et al., 2003).

This is consistent with the design-oriented conclusion reached by Tay et al. (2022) in their systematic review of DGBL for professional upskilling: “features in digital games that increase the cognitive load on the players without connecting to the learning objectives can be viewed as extraneous” and can “distract the players from the initial intent of the game.” Tay

et al. further note that longer time spent in a game and increased subjective engagement do not automatically equate to better learning outcomes (citing Dankbaar et al., 2016; Proske et al., 2014), particularly when learners are situated in highly complex simulations before foundational schemas are in place. The present study’s CL decomposition—equivalent directed effort but substantially higher perceived difficulty and complexity—maps precisely onto this distinction: the game’s card mechanics and scenario structure added interface burden that did not translate into productive cognitive engagement, especially for novice learners.

The result does not mean that CLT was irrelevant to the intervention; rather, the theory diagnosed the problem more successfully than the design solved it. The game was built on a sensible assumption that segmentation, staged progression, and immediate feedback would reduce unnecessary load. The empirical pattern indicates that these features did not sufficiently offset the novelty of the mechanics for the mix of learners who actually participated, particularly those at the novice end of the expertise distribution.

6.3 Self-Efficacy, Engagement, and Transfer

Within the game group, self-efficacy, engagement, transfer intention, and adaptive selling were all strongly interrelated, indicating that the motivational side of the theoretical framework was functioning coherently. Learners who felt more capable also felt more engaged, more willing to transfer their learning, and more adaptive in their perceived selling approach—consistent with Bandura’s (1977, 1986) view that efficacy beliefs are closely tied to persistence, investment, and anticipated future action. Critically, mediation analysis showed that SE *fully* mediated the effect of engagement on adaptive selling (indirect effect = 1.10, 95% CI [.43, 2.24]), with the direct engagement–ADAPTS path becoming non-significant once SE was included. The game’s primary internal learning mechanism therefore appears to operate through confidence-building: engagement helps only insofar as it first builds self-efficacy, which then drives adaptive selling.

The Fisher z moderation results extend this interpretation: the game significantly tightened the SE–ADAPTS coupling relative to the control condition ($r = .877$ vs. $.532$, $p = .004$). In the control group, self-efficacy and adaptive selling are correlated but more loosely integrated; in the game group they become nearly synonymous. This suggests that the game’s iterative scenario-based structure, when it works, creates tightly reinforcing loops between confidence and perceived adaptive capability.

The finding that SE fully mediated the engagement–ADAPTS pathway—with the direct engagement–ADAPTS path becoming non-significant once SE was controlled—is consistent with a broader pattern noted in the upskilling literature. Tay et al. (2022) observe that interest in a game “is the strongest factor of fantasy in educational games, but it did not show a significant effect on academic achievements” in isolation. Engagement and motivation are necessary but not sufficient: they require an intervening confidence-building mechanism to produce learning outcomes. In our data, that mechanism is SE. This points to a specific design implication: game features that generate engagement without also generating mastery experiences will not produce learning gains, regardless of how motivating they feel.

The absence of a meaningful relationship between cognitive load and motivational outcomes ($CL \leftrightarrow SE$ $r = .043$; $CL \leftrightarrow ENG$ $r = .236$, both ns) suggests that subjective load captured a complicated experience rather than straightforward overload. Some learners appear to have experienced the game as effortful but worthwhile, whereas others encountered difficulty without a corresponding gain in confidence. The analyzed game sample, after substantial attrition, was also restricted to more persistent participants, which would attenuate observable load–motivation relationships. The between-group result nonetheless stands: the game imposed greater perceived difficulty without producing superior outcomes at the aggregate level.

6.4 Expertise-Conditional Effects and Theory Extension

The aggregate findings become substantially more interpretable when decomposed by learner expertise. The exploratory cluster solution identified three profiles differing primarily in pre-training SE and ADAPTS: novice learners with weak prior schemas (C1, $n = 10$), moderate-experience learners (C2, $n = 28$), and advanced learners with strong prior schemas (C3, $n = 25$). The within-stratum game vs. control comparison on ADAPTS gain shows a monotone expertise reversal pattern: game severely disadvantages novices ($d = -1.88$), converges with control for moderate learners ($d = -0.15$), and shows a small non-significant edge for advanced learners ($d = +0.24$).

This pattern is consistent with expertise reversal theory (Kalyuga et al., 2003): instructional supports that benefit low-knowledge learners in principle can become extraneous overhead—or worse, can actively interfere—when learners possess schemas sufficient to process the

domain without guided scaffolding. For game-based training, the implication is that mechanics which are motivating for experienced learners (card strategy, scenario challenge, token management) are precisely the mechanics that overwhelm novices still acquiring basic domain knowledge.

The CLT mechanism is specific: for novice learners, working memory must simultaneously handle unfamiliar game mechanics *and* novel sales concepts. With schema space occupied by mechanic learning, domain encoding is crowded out. For experienced learners, mechanics activate existing schemas with little extraneous overhead, enabling the game’s interactive affordances to serve learning rather than obstruct it. The behavioral data corroborate this interpretation: novice game learners were slower in phase 1, less accurate, and reported markedly lower engagement and transfer intention than control-arm novices ($d = -2.06$ and $d = -1.46$, respectively).

These cluster results are exploratory and subject to the caveats noted in Section 5.6: the solution was not pre-registered, cluster stability was not formally tested across seeds, and survivor bias in the novice game cluster means the true disadvantage may be larger than observed. Nonetheless, the magnitude and theoretical coherence of the pattern strengthen the case for treating expertise as a key moderator in future GBL research and deployment decisions.

The expertise-conditional pattern is not without precedent in the organizational GBL literature. Hernández and Moreno’s (2019) systematic review of 60 organizational training GBL studies identified a recurring asymmetry: among learners with high prior experience and positive attitudes toward games, GBL produced better outcomes than traditional methods, while among learners with low prior experience and attitudes, GBL produced worse outcomes than conventional training. Although their review characterises the moderator as gaming experience rather than domain expertise, the structural pattern is identical: the same interactive design that benefits prepared learners actively disadvantages unprepared ones. The present study provides a controlled, individual-participant-level test of this asymmetry in a professional sales training context, offering more precise causal leverage than the review-level observation alone.

The conditional pattern also aligns with moderation evidence from Liu and Zhou’s (2025) professional training RCT, which found that gamified training had a significant indirect effect on knowledge and attitudes but no significant effect on skills ($t = 1.53$, $p = .131$). That skills — the most procedural and transfer-demanding outcome — resisted gamification effects in their otherwise positive trial contextualises our ADAPTS null finding: even when

gamified training works motivationally, skill-level competency gains are the hardest outcome to move. Their study further showed that the cognitive pathway from gamified training to performance depended critically on trainee need for achievement (high NFA: $\hat{\beta} = 0.51$, $p < .001$; low NFA: $\hat{\beta} = -0.01$, ns), a structural parallel to our finding that game benefit depends on prior domain expertise. Together, these studies suggest a general principle: gamified training activates cognitive and motivational pathways conditionally, with trainee entry characteristics serving as the key boundary condition.

6.5 Linkback to the Theoretical Framework

Viewed against the conceptual framework, the evidence is expertise-conditional rather than uniformly positive or uniformly negative. The CLT component is supported at the efficiency level: the intervention that was supposed to manage load produced more of it, especially for novices. The self-efficacy and engagement components are supported at the correlational and mediation level: among completers in the game condition, SE fully mediated the engagement–ADAPTS pathway, and the SE–ADAPTS coupling was significantly stronger than in the control condition. The expertise reversal pattern provides a theoretically coherent account of *why* aggregate effects were marginal: the beneficial and harmful sub-effects cancelled in the aggregate.

The LM-GM logic is partly realized. The intervention mapped mechanics to intended learning functions—classification tasks for identification, scenario encounters for application, feedback for reflection and mastery. For learners with sufficient prior schemas (C2, C3), this alignment appears to have worked: mechanics activated rather than competed with domain knowledge. For novice learners (C1), the same mechanics first had to be learned before learning through them could begin, consuming working memory capacity that should have been available for domain encoding.

Arnab et al.’s (2015) own Bloom’s taxonomy mapping of game mechanics to learning functions (their Figure 3) makes the structural reason for this failure visible in retrospect. The Phase 1 mechanics at the core of the game—tokens, selecting/collecting, cascading information, and Pavlovian interaction feedback—map to *Retention*-level learning functions (Repetition, Instruction, Generalisation). Phase 2’s role-play and simulation mechanics sit at the *Applying* and *Understanding* levels. The LM-GM alignment was therefore not only structurally correct but Bloom-level coherent: the intended learning trajectory moved from lower-order retention through to application, exactly as the framework recommends. What

the framework does not account for is that Retention-level mechanics must be decoded and automated before the Applying-level mechanics become accessible—a Bloom-level prerequisite dependency that is invisible at the design stage but directly observed in our data. Novice learners, who lacked both domain schemas and game-decoding experience, were effectively trapped at the Retention layer: the Phase 1 mechanics consumed their available cognitive resources before the Phase 2 Applying-level learning could begin.

This is the key theoretical contribution of the study. The results do not show that DGBL is incompatible with sales training, nor that the LM-GM framework fails in principle. They show that both the game’s learning benefits and the framework’s structural logic are expertise-conditional: CLT, self-efficacy theory, and LM-GM together predict when game mechanics will serve learning, but only when the prior-knowledge assumption underlying each theory is met. The design failed not because theory was irrelevant, but because the framework’s alignment guarantee is a design-level guarantee that does not extend to the cognitive preconditions required for learners to activate it. Accounting for those preconditions—mapping not only the LM-GM structure but the mechanics transparency cost and its Bloom-level prerequisite dependencies—is the extension to the framework that this study motivates.

6.6 The Importance of Non-Completers

A central contribution of this manuscript is to keep non-completers visible rather than collapsing the study into the final analyzed sample alone. In online intervention studies, attrition is often partially caused by the design itself. Here, the game arm lost more than half of those allocated to it before analysis, whereas the control arm lost roughly one fifth. This matters for both internal and applied interpretation.

Methodologically, the final comparison is per-protocol and therefore reflects the experience of completers rather than the experience of all randomized participants. Practically, the attrition suggests that the game format presented meaningful barriers to sustained participation. Because the dropouts were not obviously lower in baseline self-efficacy, the issue does not appear to be simple ability sorting. Instead, it is plausible that some participants found the game too effortful, too unclear, or too demanding relative to expectations for a short online training session. In an organizational training context, such friction would be a serious design concern.

6.7 Implications for B2B Sales Training Design

The expertise-conditional findings point to a specific practical design recommendation: route learners by prior knowledge before assigning them to game-based training. The pre-training SE and ADAPTS measures in this study reliably predicted cluster membership; a short pre-training assessment could serve as a routing mechanism. Novice learners (low pre-SE, low ADAPTS) would be directed to direct instruction first, with game-based modules introduced once foundational schemas are established. Intermediate and advanced learners could be routed directly to game-based training, where the interactive mechanics appear beneficial or at least neutral.

The pre-ADAPTS score in particular provides a deployable screening threshold. ROC analysis identified a cut-off of $\leq 46.7\%$ on the pre-intervention ADAPTS score as the optimal predictor of C1 (novice) cluster membership, achieving $AUC = 0.985$, sensitivity = 1.000, and specificity = 0.925 against the held-out completer sample. A combined pre-ADAPTS + pre-SE classifier yielded $AUC = 0.996$. While these estimates should be treated with caution given the small C1 stratum ($n = 10$), the threshold provides a concrete and measurable screening criterion: practitioners administering a standard pre-training ADAPTS-SV assessment could flag participants below this score for direct-instruction routing before game enrolment, requiring no additional instruments or administrative infrastructure.

For design practice, the results reinforce a principle of progressive complexity: early modules for novices should reduce interface ambiguity, provide worked-example guidance, and separate mechanic onboarding from conceptual instruction. Later modules, or adaptations for experienced learners, can introduce strategic decision-making, branching scenarios, and open-ended challenges. Dynamic difficulty adjustment—recognising when a learner’s accuracy or response time signals overload and reducing mechanic complexity—could serve as an automated version of expertise routing.

More broadly, these findings suggest that DGBL should be treated as a design hypothesis that must earn its value through empirical calibration across learner expertise levels. Aggregate engagement benefits are not sufficient justification for deployment if the same design systematically disadvantages a significant portion of the target population. In theory-consistent terms: mechanics should remain subordinate to learning mechanics, scaffolding must anticipate the full range of prior knowledge in the deployment population, and feedback must be interpretable enough to generate mastery experiences rather than confusion.

6.8 Limitations

Several limitations should be acknowledged. First, the final analyzed sample was modest ($N = 63$), and the large post-randomization attrition—especially in the game condition (41.8% vs. 16.2%)—limits generalizability and means the completer comparison is not equivalent to an intention-to-treat estimate. Post-hoc power analysis (Table 6.1) confirms that the study was adequately powered only for the cognitive load comparison; the two primary outcomes were substantially underpowered. The CL comparison (power = 0.792) is likely to replicate; the SE finding (power = 0.462) and ADAPTS null (power = 0.313) should be treated as preliminary until replicated in larger samples.

Table 6.1: Retrospective power analysis: required N and achieved power for key outcomes

Outcome	Obs. η_p^2 / d	N	α	Power achieved	N for 80% power
Self-efficacy (SE)	.056	63	.05	0.462	~140
ADAPTS	.036	63	.05	0.313	~220
Cognitive load	$d = 0.71$	63	.05	0.792	~66

Note. Power computed using G^* Power conventions: ANCOVA for SE and ADAPTS (covariate ρ estimated from observed pre-post correlation); independent-samples t -test (two-tailed) for CL. $\alpha = .05$ throughout.

Second, while behavioral log evidence supports a Missing At Random (time-tolerance threshold) interpretation of attrition, this cannot be confirmed without follow-up data from non-completers. A further interpretive caveat applies to the behavioral telemetry more broadly: as Lim et al. (2021) demonstrate in an online learning context, platform trace data capture only the observable portion of a learner’s behavior and do not represent the full range of self-regulatory activity that may occur outside the tracked environment. The first-attempt latency and phase-1 accuracy measures used here should therefore be treated as indicators of in-game processing rather than comprehensive indices of learning engagement. Examining dropout timing adds nuance: participants who exited within 5 minutes of starting (before completing phase 1 content) are more plausibly responding to cognitive friction and task ambiguity than to time unavailability, since 5 minutes is insufficient for a genuine scheduling conflict to resolve. The MAR assumption is strongest for the 5–15 minute dropout band, where participants had engaged with substantive content before discontinuing; these dropouts are most consistent with a time-tolerance-based decision. Participants in the >15-minute dropout band had completed a substantial proportion of the task, and their non-

completion is more puzzling; if their pre-survey data become available through follow-up, inclusion in a prospective ITT-B analysis would be informative.

Third, the exploratory cluster analysis was not pre-registered, and cluster stability across different random seeds was not formally tested. The final cluster solution used two pre-intervention features only (Pre-SE and Pre-ADAPTS), yielding silhouette = 0.49 — acceptable but not strong separation. This pre-intervention-only approach was adopted to avoid circularity: an earlier exploratory version of the analysis used 10 features including post-test and gain variables (silhouette = 0.26), which risks partially defining clusters by the very outcomes used to evaluate them; that version was replaced in the final analysis. The expertise-reversal finding ($d = -1.88$ for novices) is theoretically compelling but should be confirmed in a pre-registered replication with a fully powered sample before being treated as definitive. A survivor-bias caveat also applies: game-arm novices who completed the study ($n = 4$) may be the most persistent subset of novice participants, meaning the true game disadvantage for novices could be larger than observed.

Fourth, all primary outcome measures are self-report. The high SE-ADAPTS correlation ($r = .877$ in the game group) is partly attributable to construct overlap—both measures tap confidence in sales tasks—and shared method variance from the same post-test session. No objective competency measure was included; all competency-related findings should be interpreted as self-report beliefs about adaptive selling, not demonstrated skill acquisition.

Fifth, the study focused on short-term post-test outcomes and perceived transfer intention, without long-term follow-up on retention or real-world skill transfer. The transfer intention measure (Tho and Trang, 2015) captures post-training intention rather than demonstrated on-the-job transfer; intention-behavior gaps are well-documented in the training transfer literature (Blume et al., 2010), and transfer intention should be interpreted accordingly as a proximal motivational indicator rather than an outcome measure.

Sixth, a format-measurement alignment concern warrants acknowledgement as an alternative explanation for the ADAPTS null finding. The control condition delivered content in propositional text slides, and the ADAPTS-SV questionnaire assesses adaptive selling beliefs through propositional, declarative statements about sales behavior (e.g., “I vary my sales style from situation to situation”). The surface format of the measurement instrument therefore more closely matches the surface format of the control intervention than the game format. If any portion of ADAPTS-SV performance reflects familiarity with propositional descriptions of selling concepts, independent of genuine belief change, the control group’s advantage may be partially attributable to this format match rather than to superior learn-

ing. This alternative explanation cannot be ruled out from the present data. Future studies comparing game-based and text-based training should consider including scenario-based or process-oriented performance measures that are format-neutral relative to both intervention arms.

Seventh, a measurement timing concern applies to comparative GBL research more broadly, and to this study in particular. Immediate post-test comparisons are structurally favourable to instructional conditions that encode propositions directly: a text-based module that presents propositional content is compared, moments later, on a measure of propositional knowledge or propositional beliefs, producing a near-identical format–timing match. Game-based learning, by contrast, operates on different psychological levers — schema formation through repeated simulation, self-efficacy building through mastery experiences, and motivational priming for continued post-training engagement. These mechanisms are not primarily propositional, and their outputs are not primarily immediate: they are likely to manifest more clearly in delayed retention, in on-the-job behavioural transfer, and in self-directed post-training learning than in immediate post-test measurement. This study attempted to partially address the problem by using psychometric belief measures (SE, ADAPTS) rather than declarative knowledge tests, avoiding the most direct format match. However, even belief measures at immediate post-test capture the propositional-encoding advantage of the control condition more directly than the game’s motivational–schema pathway. Empirical support for this temporal asymmetry comes from Wouters et al.’s (2013) meta-analysis, which found that the serious game advantage was larger for *retention* ($d = 0.36$) than for *immediate learning* ($d = 0.29$) — a pattern consistent with games producing learning through mechanisms that compound over time rather than being fully expressed at the moment of training completion. The SE-mediated engagement pathway observed within our game group is consistent with precisely this mechanism: the game built confidence and motivational priming that, measured at a delayed follow-up, would be predicted to generate continued self-directed learning and behavioural change. We captured the mechanism but not its product. Future studies in this space should treat delayed measurement — at a minimum 4–6 weeks post-training, with an on-the-job transfer indicator — as a primary rather than supplementary endpoint, and should frame immediate post-test comparisons accordingly as conservative lower-bound estimates of game effectiveness rather than definitive outcome assessments.

Eighth, the sample composition limits generalizability. The study recruited participants from a single professional development context, with approximately 69% working in the education sector and the remainder distributed across industries, predominantly from Southeast Asia, in a voluntary online enrolment setting. This sample profile is unlikely to be representative

of B2B sales professionals in other sectors, regions, or organizational training contexts. The expertise distribution (with a relatively small novice stratum, $n = 10$) may also differ from the target population in deployed training contexts, where novices are often the primary intended audience.

Finally, the study was conducted in a web-based self-enrolment environment, which improves ecological realism for digital training but reduces control over attention and context. Regarding the comparison group: the control module included genuine learner agency — non-linear menu navigation, content skipping for familiar material, and a Phase 2 scenario component structurally parallel to the game’s Phase 2. The heterogeneous pacing observed in behavioral logs (39% of control participants with sub-10-second inter-slide gaps) is consistent with agentic selective review rather than disengagement, and should not be interpreted as evidence that the control was a minimal-effort baseline. If anything, this design makes the comparison more ecologically valid; future studies might nonetheless add explicit measures of within-module navigation behaviour to characterise control condition engagement more precisely.

6.9 Future Research

The most important next step is a pre-registered replication that stratifies participants by prior sales knowledge before randomisation, enabling a confirmatory test of the expertise-conditional hypothesis. A sufficiently powered study (approximately $n = 120$ for 80% power at $\eta_p^2 = .056$) with formal expertise moderator analysis—replacing the post-hoc cluster approach with a pre-registered continuous moderation model (pre-ADAPTS $\times$ condition interaction)—would provide stronger evidence about for whom game-based sales training works and under what conditions.

Future designs should also test redesigned game versions that separate interface onboarding from instructional content, with a dedicated non-scored tutorial phase before domain-relevant gameplay begins. Dynamic difficulty adjustment mechanisms—triggered by phase-1 accuracy or response time thresholds—could be tested as a way to route novice learners toward more supported instruction without abandoning the game format entirely.

Longer-term follow-up is needed to determine whether the expertise-conditional pattern persists, attenuates, or reverses over time. More fundamentally, delayed measurement should be treated as a primary endpoint in future GBL comparative trials. Immediate post-test com-

parisons are structurally conservative for game conditions: games operate through schema-building, confidence priming, and motivational carry-forward mechanisms whose outputs compound post-training rather than fully expressing at the moment of completion. A 4–6 week follow-up with behavioural transfer indicators and retention measures would provide a more ecologically valid test of whether the game’s motivational architecture translates into durable learning advantage, and would directly test the delayed-effect prediction that our SE-mediation finding implies. Finally, adopting validated multidimensional cognitive load instruments (e.g., Leppink et al.’s 2013 intrinsic/extraneous/germane subscale) in future studies would allow cleaner decomposition of the load types that mediate expertise-conditional effects, moving the field beyond the exploratory decomposition reported here.

Chapter 7

Conclusion

This thesis evaluated a scaffolded digital game-based intervention for B2B sales training against a conventional digital module in an online randomised controlled trial. The aggregate evidence does not support the claim that the game format produced uniformly superior learning outcomes: ANCOVA revealed only a marginal trend favouring the control group on self-efficacy ($F(1,60) = 3.56$, $p = .064^\dagger$, $\eta_p^2 = .056$) with no significant ADAPTS difference, while the game imposed substantially higher cognitive load ($d = 0.71$). But this aggregate picture is incomplete.

Exploratory cluster analysis revealed that the near-null aggregate conceals a strong expertise-conditional pattern: novice learners were markedly disadvantaged by the game ($d = -1.88$ on ADAPTS gain), while intermediate and advanced learners were neutral or marginally benefited. The mechanism is consistent with CLT and expertise reversal theory—game mechanics impose extraneous load before domain schemas exist for novices, while activating and reinforcing those schemas for experienced learners. Within game completers, SE fully mediated engagement effects on adaptive selling, and the game significantly tightened SE–ADAPTS coupling ($r = .877$ vs. $.532$), showing that the theoretical learning mechanisms were present among those who persisted.

The practical lesson is therefore more specific than “game-based training works” or “it doesn’t”: game-based B2B sales simulation is expertise-conditional. Deployed without expertise routing, it will systematically disadvantage its most vulnerable population of learners while offering limited benefit to those who least need it. Deployed to learners with sufficient prior schema, it shows potential to create tighter, more integrated learning outcomes than conventional instruction.

The cognitive load decomposition sharpens this conclusion considerably. The game elevated perceived difficulty and complexity (CL2+CL3, $d = 0.90$) without a corresponding increase in directed mental effort (CL1, $d = 0.13$, ns) — the CLT signature of extraneous rather than germane load. This dissociation is best understood through the lens of *cognitive load as currency*. Every unit of working memory capacity consumed by game mechanics — learning the card system, token economy, phase structure, and scoring logic — represents a transaction cost that must be recouped through downstream learning gains. Motivational and engagement mechanics can offset this cost: when they generate mastery experiences, they convert the overhead into germane processing and schema formation. Among completers in the game condition, the SE-mediated engagement pathway ($\beta_{\text{indirect}} = 1.10$) suggests the conversion occurred for those who persisted long enough to reach it. For novice learners, the evidence tells a different story: extraneous overhead accumulated, self-efficacy eroded, and the motivational mechanisms that might have offset the cost were blocked before they could activate. The cognitive load was paid; the learning dividend was not returned. Crucially, the intrinsic load of the domain content was held constant across conditions — both groups encountered identical sales concepts — meaning the elevated load in the game condition is attributable almost entirely to the mechanics overhead itself, not to any increase in content complexity.

This pattern exposes a structural gap in the LM-GM framework as currently articulated. Arnab et al. (2015) and Plass, Homer and Kinzer (2015) specify that game mechanics should be designed to serve learning mechanics — each game mechanic mapping onto an intended learning function, so that domain learning is embedded in gameplay rather than appended to it. The present design followed this principle: classification tasks mapped to identification learning, scenario encounters to application practice, feedback loops to reflection and mastery. Yet the empirical outcome reveals that functional alignment at the design level is necessary but not sufficient.

Importantly, this failure mode is one that Arnab et al. (2015) themselves foreground in their own case study application of the framework. Analysing *Re-Mission*, they note that “in several SGs, extraneous (i.e. pedagogy-independent) game mechanics are often designed to enhance game play. Consequently, learning occurs only tangentially, and mainly due to the contents.” The LM-GM framework was developed precisely to prevent this — to ensure each mechanic earns its place by serving a learning function. But the present study reveals a second, subtler failure mode that the framework does not address: mechanics that are pedagogically aligned at the design level can still become pedagogically independent in practice, if the learner lacks the cognitive resources to decode the mechanic as meaningful

before the learning opportunity expires.

A further structural limitation becomes visible when the game mechanics are mapped to Arnab et al.’s own Bloom’s taxonomy classification (their Figure 3). The Phase 1 mechanics central to this game — tokens, selecting/collecting, cascading information, and Pavlovian interaction feedback — map to *Retention*-level learning functions in Arnab’s framework. But adaptive selling knowledge — recognising customer signals, generalising selling strategies across contexts — sits at the *Applying* and *Understanding* levels of the same taxonomy. The Phase 1 Retention-level mechanics must therefore be mastered before Phase 2’s higher-order mechanics (simulation, role-play, scenario response) can become accessible. Arnab et al. themselves note that low-level game mechanics focus on “inter/action type game mechanics that have little pedagogical value” in isolation; their pedagogical value only emerges when they are subordinated to higher-level goals — which requires the player to have already decoded and automated them. For experienced learners, this Bloom-level prerequisite is satisfied on entry: the mechanics are rapidly decoded and immediately subordinated to domain strategy. For novices, the prerequisite creates an unintended bottleneck: Phase 1 Retention mechanics consume the working memory capacity that should be directed at the Applying-level learning the game was ultimately designed to produce.

The framework describes a structural mapping; it does not account for the temporal, sequential cost of that mapping, nor for the Bloom-level prerequisite dependency it can create. Game mechanics must first *be learned* before they can serve as scaffolds for domain learning. For learners with sufficient prior schemas, mechanic onboarding is rapid and cognitively cheap; the aligned mechanics immediately activate domain knowledge and the framework performs as intended. For novices, onboarding consumes precisely the working memory capacity that should be directed at domain content, and the alignment becomes a design fiction — the mechanic serves its intended learning function in principle, but never reaches that function in practice because the learner’s cognitive resources are exhausted by the interface before domain processing can begin. The LM-GM framework, in other words, implicitly assumes a *mechanics transparency* that is expertise-dependent and cannot be verified at the design stage alone.

Converging evidence from adjacent frameworks corroborates this reading and points toward a theoretical extension. Kirschner, Sweller and Clark (2006) argue that minimally guided instruction fails novices because simultaneous processing of unfamiliar structure and unfamiliar content saturates working memory; although their critique targets discovery learning, the mechanism is identical here — the game is not minimally guided on domain content, but it is effectively minimally guided on its own mechanics. Novice participants thus faced

dual novelty: unfamiliar sales concepts and unfamiliar game mechanics competing for the same limited cognitive resource pool. Mayer’s (2009) coherence principle provides a complementary account: interface elements that do not directly support the current learning objective introduce extraneous processing, and for learners who lack the schemas to interpret those elements as meaningful, the interface becomes noise rather than scaffold. The token economy and multi-phase structure — motivating for experienced players who can subordinate mechanics to strategy — may function as seductive details for novices still acquiring domain vocabulary. Paas and van Merriënboer’s (1994) efficiency framework offers a final operationalisation: instructional efficiency combines performance gain with invested mental effort, and a design that elevates effort (extraneous overhead) without commensurate performance gain is, by definition, inefficient for that population. Together, these frameworks converge on a design criterion that LM-GM does not yet operationalise: a *mechanics transparency coefficient* — the working memory overhead required to operate the game interface, independent of domain content. This coefficient sets a lower bound on the learner expertise at which a given game design can be expected to produce net learning benefits, and it must be calibrated empirically for the actual deployment population rather than assumed from the design intent alone.

The contributions of this study can therefore be stated precisely across three levels.

Theoretically, the study extends expertise reversal theory into professional digital GBL and identifies two design criteria that the LM-GM framework does not yet operationalise. First, a *mechanics transparency coefficient* — the working memory overhead required to operate a game interface, independent of domain content — sets a lower bound on the learner expertise at which aligned mechanics can generate net learning benefits. Alignment at the design level is necessary but not sufficient; mechanics must also be transparent enough for the target population to reach the activation threshold at which alignment performs its intended function. Second, a *Bloom-level prerequisite dependency* can emerge when the game mechanics required to access higher-order learning objectives (Applying, Understanding) are themselves Retention-level tasks that must be mastered before domain processing can begin. For experienced learners, onboarding is rapid and cognitively cheap; for novices, it consumes exactly the working memory capacity that was intended for domain learning. The cognitive load decomposition frames this through the lens of *CL-as-currency*: every unit of extraneous overhead is a transaction cost that must be recouped through downstream learning gains, and the SE-mediated engagement pathway ($\beta_{\text{indirect}} = 1.10$) shows that the conversion occurred among completers who persisted long enough to reach it. Self-efficacy, not game format per se, is the proximal mechanism linking engagement to adaptive selling beliefs — a mediating

pathway whose detection required the controlled conditions of this RCT.

Practically, the expertise-conditional finding reframes the deployment question. The aggregate null on ADAPTS is not a verdict against game-based sales training; it is a warning against undifferentiated deployment to mixed-expertise audiences. For practitioners, the actionable prescription is expertise routing before enrolment: the pre-ADAPTS threshold of $\leq 46.7\%$ correctly identified novice cluster membership with $AUC = 0.985$, sensitivity = 1.000, and specificity = 0.925, and is deployable with the same pre-training survey already required for adaptive sales training administration. Dropout timing in behavioral logs adds a complementary signal: exits within 5 minutes indicate cognitive friction rather than scheduling constraints and should trigger real-time routing or tutorial scaffolding, not passive re-invitation. Together, screening threshold and attrition monitoring provide an operationalisable deployment standard that requires no additional instrumentation beyond what this design already employed.

Methodologically, the study demonstrates three practices that the GBL evaluation literature should adopt more broadly. Pre-intervention-only clustering — restricting the segmentation features to pre-training data and validating stability across 100 random seeds ($ARI = 1.000$) — provides a template for identifying moderation-relevant subgroups without circularity. Pairing a pre-registered continuous moderation analysis (pre-outcome $\times$ condition interaction) as the primary evidence for expertise effects, rather than treating post-hoc cluster analysis as confirmatory, strengthens causal inference without requiring pre-registration of the cluster structure itself. Two ITT sensitivity analyses (no-gain imputation and worst-case) bound the direction of attrition bias and demonstrate that the SE finding is robust across missing data assumptions. Finally, the study contributes a measurement-timing critique that extends beyond this study to GBL evaluation research as a field: immediate post-test comparison is structurally conservative for game-based conditions, because games operate through schema-building, confidence priming, and motivational carry-forward mechanisms whose outputs compound post-training rather than fully expressing at the moment of completion. Framing immediate post-test results as conservative lower-bound estimates — and treating delayed measurement as a primary endpoint — is a methodological standard that comparative GBL research should now adopt by default.

The expertise-conditional pattern observed here is not, ultimately, a failure of game-based learning. It is a diagnostic: a quantified signal of where the mechanics transparency threshold lies, what the Bloom-level prerequisite dependency costs novice learners, and what a next-generation game design would need to change to recoup that cost. The study answers its central question — *for whom does game-based learning work?* — not with a blanket verdict

but with a precision prescription: for learners who arrive with sufficient prior schema to decode the mechanics as meaningful, and who are routed in before the extraneous overhead accumulates into a barrier.

Bibliography

- [1] All, A., Castellar, E. N. P., and Van Looy, J. (2021). Digital game-based learning effectiveness assessment: Reflections on study design. *Computers & Education*, 167, 104160.
- [2] Arnab, S., Lim, T., Carvalho, M. B., Bellotti, F., de Freitas, S., Louchart, S., Suttie, N., Berta, R., and De Gloria, A. (2015). Mapping learning and game mechanics for serious games analysis. *British Journal of Educational Technology*, 46(2), 391–411.
- [3] Plass, J. L., Homer, B. D., and Kinzer, C. K. (2015). Foundations of game-based learning. *Educational Psychologist*, 50(4), 258–283.
- [4] Blume, B. D., Ford, J. K., Baldwin, T. T., and Huang, J. L. (2010). Transfer of training: A meta-analytic review. *Journal of Management*, 36(4), 1065–1105.
- [5] Bandura, A. (1977). Self-efficacy: Toward a unifying theory of behavioral change. *Psychological Review*, 84(2), 191–215.
- [6] Bandura, A. (1986). *Social Foundations of Thought and Action: A Social Cognitive Theory*. Prentice Hall.
- [7] Barling, J., and Beattie, R. (1983). Self-efficacy beliefs and sales performance. *Journal of Organizational Behavior Management*, 5(1), 41–51.
- [8] Bussière, D. (2017). Understanding the sales process by selling. *Marketing Education Review*, 27(2), 86–91.
- [9] Cardinali, S., Giovannetti, M., Hautamäki, P., and Burgdorff Jensen, K. (2025). A competency-based curriculum content design for complex B2B sales in higher education institutions. *Journal of Marketing Education*. Advance online publication.

- [10] Chang, C.-C., Warden, C. A., Liang, C., and Lin, G.-Y. (2018). Effects of digital game-based learning on achievement, flow and overall cognitive load. *Australasian Journal of Educational Technology*, 34(4).
- [11] Chang, C.-Y., Kao, C.-H., Hwang, G.-J., and Lin, F.-H. (2020). From experiencing to critical thinking: A contextual game-based learning approach to improving nursing students' performance in electrocardiogram training. *Educational Technology Research and Development*, 68(3), 1225–1245.
- [12] Cummins, S., Peltier, J. W., Erffmeyer, R., and Whalen, J. (2013). A critical review of the literature for sales educators. *Journal of Marketing Education*, 35(1), 68–78.
- [13] Dahalan, F., Alias, N., and Shaharom, M. S. N. (2024). Gamification and game-based learning for vocational education and training: A systematic literature review. *Education and Information Technologies*, 29(2), 1279–1317.
- [14] Evans, P., Vansteenkiste, M., Parker, P., Kingsford-Smith, A., and Zhou, S. (2024). Cognitive load theory and its relationships with motivation: A self-determination theory perspective. *Educational Psychology Review*, 36(1), 7.
- [15] Fredricks, J. A., Blumenfeld, P. C., and Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, 74(1), 59–109.
- [16] Hainey, T., Connolly, T. M., Boyle, E. A., Wilson, A., and Razak, A. (2016). A systematic literature review of games-based learning empirical evidence in primary education. *Computers & Education*, 102, 202–223.
- [17] Hernández, M., and Moreno, J. (2019). A systematic literature review on organizational training using game-based learning. In V. Agredo-Delgado and P. H. Ruiz (Eds.), *Human-Computer Interaction* (pp. 1–18). Springer.
- [18] Hung, C.-M., Huang, I., and Hwang, G.-J. (2014). Effects of digital game-based learning on students' self-efficacy, motivation, anxiety, and achievements in learning mathematics. *Journal of Computers in Education*, 1(2), 151–166.
- [19] Kalyuga, S., Ayres, P., Chandler, P., and Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, 38(1), 23–31.
- [20] Kirschner, P. A., Sweller, J., and Clark, R. E. (2006). Why minimal guidance during instruction does not work: An analysis of the failure of constructivist, discovery, problem-based, experiential, and inquiry-based teaching. *Educational Psychologist*, 41(2), 75–86.

- [21] Knight, P., Mich, C. C., and Manion, M. T. (2014). The role of self-efficacy in sales education. *Journal of Marketing Education*, 36(2), 156–168.
- [22] Kordaki, M., and Gousiou, A. (2017). Digital card games in education: A ten-year systematic review. *Computers & Education*, 109, 122–161.
- [23] Paas, F. G. W. C., and van Merriënboer, J. J. G. (1994). Instructional control of cognitive load in the training of complex cognitive tasks. *Educational Psychology Review*, 6(4), 351–371.
- [24] Paas, F., Tuovinen, J. E., Tabbers, H., and Van Gerven, P. W. M. (2003). Cognitive load measurement as a means to advance cognitive load theory. *Educational Psychologist*, 38(1), 63–71.
- [25] Mayer, R. E. (2009). *Multimedia Learning* (2nd ed.). Cambridge University Press.
- [26] Moncrief, W. C., and Marshall, G. W. (2005). The evolution of the seven steps of selling. *Industrial Marketing Management*, 34(1), 13–22.
- [27] Noroozi, O., Dehghanzadeh, H., and Talaei, E. (2020). A systematic review on the impacts of game-based learning on argumentation skills. *Entertainment Computing*, 35, 100369.
- [28] Reeve, J. (2012). A self-determination theory perspective on student engagement. In S. L. Christenson, A. L. Reschly, and C. Wylie (Eds.), *Handbook of Research on Student Engagement* (pp. 149–172). Springer.
- [29] Senderek, R., Brenken, B., and Stich, V. (2015). The implementation of game-based learning as part of corporate competence development. In *2015 International Conference on Interactive Collaborative and Blended Learning* (pp. 44–51).
- [30] Sweller, J., Ayres, P., and Kalyuga, S. (2011). *Cognitive Load Theory*. Springer.
- [31] Tay, J., Goh, Y. M., Safiena, S., and Bound, H. (2022). Designing digital game-based learning for professional upskilling: A systematic literature review. *Computers & Education*, 184, 104518.
- [32] Tho, N. D., and Trang, N. T. M. (2015). Can knowledge be transferred from business schools to business organizations through in-service training students? SEM and fsQCA findings. *Journal of Business Research*, 68(6), 1332–1340.

- [33] Woo, J.-C. (2014). Digital game-based learning supports student motivation, cognitive success, and performance outcomes. *Journal of Educational Technology & Society*, 17(3), 291–307.
- [34] Bösche, W., and Kattner, F. (2011). Fear of (serious) digital games and game-based learning?: Causes, consequences and a possible countermeasure. *International Journal of Game-Based Learning*, 1(3), 1–15.
- [35] Chen, Z.-H., Chen, H. H.-J., and Dai, W.-J. (2018). Using narrative-based contextual games to enhance language learning. *Journal of Educational Technology & Society*, 21(3), 186–198.
- [36] Christenson, S. L., Reschly, A. L., and Wylie, C. (Eds.). (2012). *Handbook of Research on Student Engagement*. Springer.
- [37] Church, A. (1975). Karl R. Popper: The logic of scientific discovery. *Journal of Symbolic Logic*, 40(3), 471–471.
- [38] Kirk, R. E. (2013). *Experimental Design: Procedures for the Behavioral Sciences* (4th ed.). Sage.
- [39] Knowles, M. S., Holton III, E. F., Swanson, R. A., and Robinson, P. A. (2020). *The Adult Learner: The Definitive Classic in Adult Education and Human Resource Development* (9th ed.). Routledge.
- [40] Liu, Y.-C., Wang, W.-T., and Huang, W.-H. (2023). The effects of game quality and cognitive loads on students’ learning performance in mobile game-based learning contexts: The case of system analysis education. *Education and Information Technologies*, 28(12), 16285–16310.
- [41] Leppink, J., Paas, F., Van der Vleuten, C. P. M., Van Gog, T., and Van Merriënboer, J. J. G. (2013). Development of an instrument for measuring different types of cognitive load. *Behavior Research Methods*, 45(4), 1058–1072.
- [42] Liu, S., and Zhou, H. (2025). Does gamified training improve training performance? A dual-pathway moderated mediation model. *Journal of Business Research*, 188, 115086.
- [43] Deci, E. L., and Ryan, R. M. (2000). The “what” and “why” of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, 11(4), 227–268.

- [44] Paas, F., Ayres, P., and Pachman, M. (2008). Assessment of cognitive load in multimedia learning: Theory, methods and applications. In J. L. Plass, R. Moreno, and R. Brünken (Eds.), *Cognitive Load Theory* (pp. 11–35). Cambridge University Press.
- [45] Shadish, W., Cook, T., and Campbell, D. (2004). *Quasi-Experimental Designs for Generalized Causal Inference*. Houghton Mifflin.
- [46] Sousa, M., and Costa, E. (2014). Game-based learning improving leadership skills. *EAI Endorsed Transactions on Serious Games*, 1(3), e2.
- [47] Titus, A. A. (1994). Competence at Work, by Lyle M. Spencer, Jr., and Signe M. Spencer. *Human Resource Development Quarterly*, 5(4), 391–395.
- [48] Lim, L. A., Dawson, S., Gašević, D., Joksimović, S., Pardo, A., Fudge, A., and Gentili, S. (2021). Impact of learning analytics feedback on self-regulated learning: Triangulating behavioural logs with students’ recall. In *Proceedings of the 11th International Learning Analytics and Knowledge Conference (LAK21)* (pp. 364–374). ACM.
- [49] Wouters, P., Van Nimwegen, C., Van Oostendorp, H., and Van der Spek, E. D. (2013). A meta-analysis of the cognitive and motivational effects of serious games. *Journal of Educational Psychology*, 105(2), 249–265.
- [50] Merchant, Z., Goetz, E. T., Cifuentes, L., Keeney-Kennicutt, W., and Davis, T. J. (2014). Effectiveness of virtual reality-based instruction on students’ learning outcomes in K-12 and higher education: A meta-analysis. *Computers & Education*, 70, 29–40.